

Relative Exponential Fixed Parts and Functional Periods: An Exponential Nori–Ayoub Theorem

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Abstract

We construct a relative category of exponential Nori motives over a smooth algebraic base and prove a relative functional-period theorem. The algebraic realization over $k = \overline{\mathbb{Q}}$ is built from the absolute de Rham augmentation of the Nori algebra and its relative module presentation; comparison, operation maps, and nonzero exponential realization are retained in the construction. For a lisse relative exponential motive, algebraic horizontal sections of its Gauss–Manin connection are induced by its maximal constant exponential submotive.

For each finite tensor-generated subcategory of the generic relative Kontsevich–Zagier exponential envelope and each $\lambda \in k^\times$, we obtain, in compatible fiber coordinates, an exact sequence

$$1 \longrightarrow G_{\text{diff}} \longrightarrow G_{\text{mot}} \longrightarrow G_{\text{const}} \longrightarrow 1.$$

The proof uses the relative Fourier comparison to control rank-one Betti constituents and the arithmetic theory of 1-motives to lift their algebraic connections. Over the actual field of constant exponential periods, the relative comparison algebra is Picard–Vessiot and its analytic evaluation is injective. The constant-field assertion is proved after comparison over $\mathbb{C}(S)$, not assumed in order to obtain that comparison. Numerical relations created by specialization and an elementary presentation of all categorical period relations are separate questions.

Contents

1	Introduction	2
1.1	The functional problem	2
1.2	Main results	3
1.3	Proof architecture and organization	4
2	Normalization conventions	4
2.1	The exponential sign	4
2.2	The lisse tensor normalization	5

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3	The coefficient system and the affine-line quotient	6
3.1	Algebraic geometric realization before the Nori quotient	6
3.2	The absolute augmentation and the relative module formula	8
3.3	The affine-line quotient	10
3.4	The algebraic exponential functor and the matching Betti fiber	12
3.5	Consequences of Virk’s realization theory	13
4	Fixed parts and extension from dense opens	14
5	Fourier fibers and ordinary geometric constituents	16
6	The arithmetic kernel and rank-one lifts	22
6.1	An optional split criterion	24
7	Compatible fibers and Galois exactness	25
7.1	Changing to Betti coordinates	26
8	Normalized solutions and extension of constants	27
9	The torsor and the exact constant-field argument	28
9.1	The actual constant-period base	28
9.2	An explicit equivariant comparison over $\mathbb{C}(S)$	29
9.3	Descent of differential simplicity and constants	30
10	Relation with Kontsevich–Zagier exponential periods	32
10.1	Arithmetic specialization	32
11	Applications and examples	33
11.1	Polynomial exponential integrals	33
11.2	Cyclotomic Gamma packets and the Rohrlich–Lang boundary	33
11.3	Nilsson– E packets	36
12	Concluding perspective	36

1 Introduction

1.1 The functional problem

For the arithmetic results we work over

$$k = \overline{\mathbb{Q}},$$

and let S be a smooth connected quasi-projective k -variety. The exponentiated category is defined over any algebraically closed subfield of $\mathbb{C}$. In this paper all algebraic realization and fixed-part statements use $k = \overline{\mathbb{Q}}$ and a scale in $k^\times$; over a larger algebraically closed ground field the same categorical statements apply with the scale in that field. We use the k -linear coefficient extension of the rational Nori categories, separately from geometric base extension.

A relative exponential period is represented schematically by a family

$$s \longmapsto \int_{\gamma_s} e^{-f_s} \omega_s,$$

where the de Rham class is computed with the twisted differential $d - df_s \wedge$ and the Betti class is a rapid-decay cycle. Exponential periods admit compatible naive, cohomological, and motivic descriptions [9, 18]. In families, the resulting functions are solutions of exponential Gauss–Manin systems and admit rapid-decay integral representations [24, 23].

For ordinary periods, Nori and Ayoub prove a functional form of the Kontsevich–Zagier period conjecture: the relative formal period algebra injects into the corresponding analytic function field. Equivalently, the relative motivic comparison torsor is governed by algebraic monodromy [32, 4, 5, 34, 7]. The aim of this paper is to establish the analogous statement for exponential motives.

Functional and numerical relations must be distinguished. The functional problem concerns identities valid on a dense open of the parameter space. The numerical problem asks whether additional relations appear after specialization to an algebraic point. The latter requires separate arithmetic input and is not addressed here.

1.2 Main results

Let $\mathrm{MN}(X)$ be the abelian category of perverse Nori motives on a quasi-projective k -variety X [26, 43, 44]. For

$$\pi_X : X \times \mathbb{A}^1 \longrightarrow X,$$

define

$$\mathrm{EM}_N(X) := \mathrm{MN}(X \times \mathbb{A}^1) / \pi_X^*[1] \mathrm{MN}(X). \quad (1.1)$$

Theorem 1.1 (Exponentiated perverse-Nori heart). *The essential image of*

$$\pi_X^*[1] : \mathrm{MN}(X) \longrightarrow \mathrm{MN}(X \times \mathbb{A}^1)$$

is a Serre subcategory. The quotient in (1.1) is the heart of the exponentiated derived coefficient system associated with $D^b(\mathrm{MN}(-))$. The derived quotient carries the six operations and the additive-convolution tensor product. On its lisse heart we use the normalized tensor product and fiber functor of Section 2. The Nori-to-Hodge realization descends to an exact faithful functor

$$R_H^{\mathrm{exp}} : \mathrm{EM}_N(X) \longrightarrow \mathrm{EM}_H(X)$$

compatible with these operations and with every nonzero exponential realization.

Fix $\lambda \in k^\times$. Let S have dimension d , let $a : S \rightarrow \mathrm{Spec} k$ be the structure morphism, and let $N \in \mathrm{EM}_N(S)$ be λ -lisse. Set

$$N^{\mathrm{fix}} := \mathrm{im} \left(a^* {}^p H^{-d}(a_* N)[d] \longrightarrow N \right). \quad (1.2)$$

Theorem 1.2 (Invariant exponential fixed part). *The λ -realization of N^{fix} is the maximal trivial differential submodule of $\rho_\lambda(N)$. The object N^{fix} is the maximal constant subobject of N .*

Let $\eta = \mathrm{Spec} k(S)$ and let $\mathcal{T}_{\mathrm{KZ}}(\eta)$ denote the generic rigid Tannakian envelope generated by elementary relative Kontsevich–Zagier exponential data.

Theorem 1.3 (Relative exponential Galois exactness). *Let $\lambda \in \overline{\mathbb{Q}}^\times$ and let $\mathcal{T}$ be a finitely generated rigid Tannakian subcategory of $\mathcal{T}_{\mathrm{KZ}}(\eta)$. Then the relative differential Galois group is normal in the motivic Galois group, and*

$$1 \longrightarrow G_{\mathrm{diff}} \longrightarrow G_{\mathrm{mot}} \longrightarrow G_{\mathrm{const}} \longrightarrow 1$$

is exact. The groups are taken in the compatible algebraic evaluation fiber coordinates of Section 7; Betti coordinates are related by the comparison conjugation in Lemma 7.5.

Theorem 1.4 (Relative exponential Nori–Ayoub theorem). *For the canonical realization $\lambda = 1$, after the algebraic relations among constant exponential periods have been imposed, the generic relative formal exponential-period map of the filtered Kontsevich–Zagier Tannakian envelope is injective. The same statement holds for every $\lambda \in \overline{\mathbb{Q}}^\times$.*

The fixed-part theorem follows from the lowest algebraic direct image. The proof of Theorem 1.3 requires control of differential characters. Sabbah’s relative Fourier–nearby-cycle comparison places their Betti local systems in the ordinary geometric-origin category; Lam–Litt then gives finite monodromy, and Huber–Wüstholz’s theory of 1-motives produces the required exponential-motivic lifts. The resulting semi-invariant subobjects yield normality by André’s criterion, and the comparison torsor is identified with the Picard–Vessiot torsor.

1.3 Proof architecture and organization

The proof is ordered so that the relative period theorem is never used to construct its own realization functor. Section 2 fixes the kernel, lisse tensor product, and fiber conventions. In Section 3 we construct the algebraic realization from the absolute Nori de Rham augmentation, prove its comparison and operation compatibilities, and pass to the affine-line quotient. Section 4 proves the algebraic fixed-part and dense-open extension statements. Sections 5 and 6 supply geometricity and rank-one lifting; the optional split criterion requires an actual rank-one comparison and is not used for the general saturation argument. Section 7 proves Galois exactness in compatible fiber coordinates. Sections 8 and 9 construct the Picard–Vessiot comparison first over $\mathbb{C}(S)$ and then prove the exact smaller constant field. Finally Sections 10 and 11 state the generic functional consequence and compute the cyclotomic Gamma example.

2 Normalization conventions

Put $k = \overline{\mathbb{Q}} \subset \mathbb{C}$. Let S/k be smooth, connected, and quasi-projective, of dimension d , and write $a : S \rightarrow \operatorname{Spec} k$ and $K = k(S)$. Unless otherwise stated, $\lambda \in k^\times$. Geometric ground field, coefficient field, de Rham field, and analytic constant field are not silently identified: algebraic connections are defined over k , the motivic categories are k -linear, and analytic comparison is taken after extension to $\mathbb{C}$.

All six operations in this paper use sheaf-compatible perverse normalizations. Derived functors are derived even when no R or L is written. A vector bundle with connection M on a smooth d -fold corresponds to a perverse-normalized object, while its unshifted de Rham complex has terms in degrees $0, \dots, d$.

2.1 The exponential sign

We distinguish the holonomic heart object from the derived kernel:

$$\mathcal{E}^f = (\mathcal{O}_S, d - df), \quad \mathcal{K}_\lambda = (\mathcal{O}_{\mathbb{A}_t^1}, d - \lambda dt), \quad \mathcal{L}_\lambda = \mathcal{K}_\lambda[-1]. \quad (2.1)$$

In left differential-operator notation, $\partial_t \cdot p = p' - \lambda p$ on $\mathcal{K}_\lambda$. A horizontal section is $e^{\lambda t}$; the dual exponential in an integral pairing is $e^{-\lambda t}$. Virk defines the sheaf-compatible derived kernel by requiring $\mathcal{L}_\lambda[1]$ to be this heart object [45, §6.6 and Proposition 6.8]. The shift and the sign are separate conventions, both of which matter. At the zero graph over a point, $i_0^* \mathcal{L}_\lambda = k$ in degree zero, whereas $i_0^* \mathcal{K}_\lambda = k[1]$. This verifies the unit and would detect an omitted kernel shift.

Let $\mathbf{Q}_S$ be the exponential quotient functor, and let $i_h : S \rightarrow S \times \mathbb{A}_t^1$ be the graph of $h \in \mathcal{O}(S)$.

Lemma 2.1 (Graph realization). *The normalized graph object*

$$\mathbf{G}_S(h) := \mathbf{Q}_S(i_{h,*}\mathbf{1}_S[d])$$

realizes the connection $\mathcal{E}^{\lambda h}$ at scale λ . In particular, the graph potential realizing $\mathcal{E}^f$ is $h = f/\lambda$.

Proof. Let $p : S \times \mathbb{A}^1 \rightarrow S$ and $\mu : S \times \mathbb{A}^1 \rightarrow \mathbb{A}^1$ be the projections. The algebraic realization constructed in Theorem 3.14 tensors with $\mu^*\mathcal{L}_\lambda$ and applies $p_!$. Since $pi_h = \text{id}_S$ and the graph is proper, projection formula reduces it to $\mathbf{1}_S[d] \otimes h^*\mathcal{L}_\lambda$. Its connection is $d - \lambda dh$. The $[-1]$ already in $\mathcal{L}_\lambda$ gives the unshifted rank-one local object under pullback; the remaining $[d]$ puts that connection in the perverse heart. There is no extra shift to remove and no change of sign. $\square$

For example, on $\mathbb{A}_s^1$ at scale one, the potential s gives $d - ds$, not $d + ds$. The two connections are not isomorphic over $k(s)$: a rational gauge between them would have logarithmic derivative 2 or -2 , whereas a rational logarithmic derivative tends to zero at infinity.

2.2 The lisse tensor normalization

Write $\star$ for the derived additive convolution over S , including the ordinary derived tensor product in the base direction. Its derived unit is $\mathbf{1}_S^{\text{exp}} := \mathbf{Q}_S(i_{0,*}\mathbf{1}_S)$. For perverse-lisse objects put

$$N \otimes_{\text{lisse}} M = (N \star M)[-d], \quad \mathbf{1}_{S,\text{lisse}} = \mathbf{1}_S^{\text{exp}}[d]. \quad (2.2)$$

The associativity, symmetry, and unit constraints are transported from the unshifted lisse normalization by $N \mapsto N[-d]$ and its inverse. This transport is part of the definition; the symmetry is not an unadjusted restriction of the derived Koszul symmetry.

Fix a direction at infinity for the absolute Betti functor. For the pairing with $e^{-\lambda t}$ choose the ray with $\theta = -\arg \lambda$, or transport an equivalent choice consistently. In this paper Ψ_∞ denotes this chosen directional functor $\Psi_{e^{i\theta}\infty}$, not a direction-free canonical fiber; see [42, Definition 3.5 and Theorem 3.6].

Proposition 2.2 (Normalized tensor and fiber). *On the lisse subcategory the operations in (2.2) realize ordinary tensor products of connections. For an algebraic point $s \in S(k)$, the normalized absolute restriction is $i_s^*N[-d]$. Its absolute rapid-decay Betti fiber, in cohomological variance, is*

$$\omega_{B,s}(N) = \Psi_\infty(\text{rat}^{\text{exp}}(i_s^*N[-d])). \quad (2.3)$$

In any strictly full rigid subquotient-closed lisse tensor subcategory this is an exact tensor fiber functor. In homological notation one must dualize consistently, rather than identify homology with cohomology.

Proof. For two connections, the sheaf-compatible derived tensor of their perverse-normalized objects is their usual tensor connection shifted by d . Shifting back by $[-d]$ therefore gives the usual tensor connection in the heart. Realization compatibility with derived convolution gives the first assertion, and reflection of the heart by exact faithful realization gives lisseness of the result. The same calculation for the unit gives $\mathbf{1}_{S,\text{lisse}}$.

A constant perverse-lisse object is $a^*C[d]$, and $i_s^*a^*C[d][-d] = C$. For a general lisse object, normalized restriction is the vector-space fiber of its connection, hence is in degree zero. Exact faithful realization reflects that concentration, and the usual fiber functor on connections is exact. Tensor compatibility follows by applying $i_s^*[-d]$ to (2.2). Finally the absolute nearby-fiber functor is exact, faithful, and tensor, with the rapid-decay comparison; these are the absolute convolution inputs of Katz and Fresán–Jossen, in the form recalled by [42, Corollary 2.17 and Theorem 3.6]. $\square$

Remark 2.3 (Fixed-scale duality). If $\epsilon(s, t) = (s, -t)$, then $\rho_\lambda \epsilon^* = \rho_{-\lambda}$. Virk’s actual duality formula is

$$\mathbb{D}\rho_\lambda = \rho_{-\lambda}\mathbb{D},$$

not same-scale commutation [45, Theorem 6.13]. On the rigid lisse subcategory the usual expression for normalized tensor dual is $\epsilon^*\mathbb{D}N(-d)$: the Tate twist removes the smooth-base orientation twist. Evaluation and coevaluation are understood in the transported normalization. In particular the tensor inverse of $G_S(h)$ is $G_S(-h)$; this follows directly from graph addition and does not rely on a bare Verdier-duality assertion. We do not infer that every object of the entire perverse heart is rigid.

3 The coefficient system and the affine-line quotient

External input 3.1 (Ordinary Nori structure and absolute comparison). We use the rational perverse-Nori coefficient system of Ivorra–Morel, Terenzi, and Tubach, with its enhanced six operations, exact faithful perverse Betti realization, and conservative derived Betti realization [26, 43, 44]. Two precise additional forms of these established inputs will be used.

First, put $\mathrm{DM}(X) = \mathrm{DM}_{\mathrm{et}}(X, \mathbb{Q})$ and $\mathrm{DN}(X) = \mathrm{Ind} D^b(\mathrm{MN}_{\mathbb{Q}}(X))$. If $\mathrm{Nor}^* \dashv \mathrm{Nor}_*$ is Nori realization and

$$\mathcal{N}_k = \mathrm{Nor}_* \mathbf{1}, \quad \mathcal{N}_X = a_X^* \mathcal{N}_k,$$

then $\mathcal{N}_X \simeq \mathrm{Nor}_{X,*} \mathbf{1}$ and

$$\mathrm{DN}(X) \simeq \mathrm{Mod}_{\mathcal{N}_X}(\mathrm{DM}(X)). \quad (3.1)$$

Under this symmetric monoidal equivalence, $\mathrm{Nor}_X^* Q$ is the free module $\mathcal{N}_X \otimes Q$. The equivalence is compatible with pullback and the constructible six operations. These are Tubach’s Lemma 4.12, Corollary 4.14, and Proposition 4.16; the compatibility is part of the construction in his §4.2.

Second, at $\mathrm{Spec} k$ ordinary Nori motives have the exact tensor de Rham fiber d_k over k , the rational Betti fiber b_k , and their natural complex comparison. Their derived versions compute the corresponding geometric realization complexes, compatibly with tensor products and comparison. This is the *absolute* mixed-realization construction, not a relative realization assumption; see the cellular construction and mixed-realization factorization in [21, Theorems 7.4.17 and 7.6.5] and [11, §§6–7, especially Propositions 7.1 and 7.12]. We use their complex-level constructions and extend them by Ind . The covariant convention here is the homological one in Choudhury–Gallauer; when using Harer’s contravariant cohomological construction, take its dual on rigid geometric motives. This fixes variance before any comparison of complexes.

The relative algebraic realization is constructed below from the stated ordinary and absolute inputs. The construction uses the *absolute* Nori algebra in (3.1); a choice of separate algebraic models for complex Hodge objects would not give this construction.

3.1 Algebraic geometric realization before the Nori quotient

Let $\mathcal{D}_k(X)$ be the Ind -completion of the bounded constructible regular-holonomic algebraic differential-module category on X/k , in sheaf-compatible normalization. For singular X , use modules supported on a local smooth ambient variety and Kashiwara equivalence. Transfer bimodules and Spencer complexes give the enhanced operations. On a smooth d -fold the unshifted tensor unit is $\mathcal{O}_X[-d]$, not the holonomic heart object $\mathcal{O}_X$.

Lemma 3.2 (Geometric de Rham coefficient system). *There is a colimit-preserving symmetric monoidal geometric realization*

$$r_X : \mathrm{DM}(X) \longrightarrow \mathcal{D}_k(X)$$

which commutes with the six operations on constructible objects. After $k \hookrightarrow \mathbb{C}$ and regular Riemann–Hilbert, it is the complexified Betti realization, as a morphism of coefficient systems.

Proof. Use algebraic differential operators and their transfer bimodules over k , rather than analytifying and choosing a descent afterward. Their standard localization, base-change, projection, and smooth-purity maps make $\mathcal{D}_k$ a rational stable coefficient system. These are the ordinary algebraic differential-module operations; a realization model retaining the filtered module over k together with its complex comparison is explicitly described in Saito’s §1.8(ii), with the operations developed in his §§2–6 [41]. No identification of all such mixed sheaves with Nori motives is asserted here.

For clarity, the motivic localization conditions can be checked directly. Etale descent is descent of the algebraic modules and transfer complexes. For the affine-line projection, the relative algebraic de Rham complex of the constant object is $\mathcal{O}_X[t] \xrightarrow{\partial_t} \mathcal{O}_X[t]$, with cohomology $\mathcal{O}_X$ in degree zero; integration of polynomials is allowed in characteristic zero. Projection formula gives $\mathrm{id} \simeq R\pi_*\pi^*$. The projective-line trace identifies the reduced projective-line object with an invertible shifted Tate line, giving stability. The usual de Rham orientation gives the rational motivic factorization. These checks yield r_X by the universal coefficient-system construction [17, Theorem 7.14]. One restricts to the geometric constructible image to obtain six-operation compatibility, using the constructible realization argument of [44, Theorem 4.5 and its proof]; initiality alone is not being used to assert compatibility with every right adjoint.

The relative de Rham theorem over $\mathbb{C}$, the trace map, and the Kunneth map identify this system with Betti realization on smooth geometric objects and their structure maps. Etale descent, affine-line localization, and Tate stabilization extend this as a natural symmetric monoidal comparison of the enhanced systems. The comparisons of adjunction and base-change maps are the ones induced by the transfer complexes. Thus the comparison is a comparison of functors and maps, not just of cohomology dimensions or individual isomorphism classes. $\square$

Lemma 3.3 (The absolute comparison in the needed form). *On the absolute enhanced categories there is a symmetric monoidal identification*

$$d_k \mathrm{Nor}_k^* \simeq r_k, \tag{3.2}$$

and its complexification agrees with the actual absolute de Rham–Betti comparison. The identification is compatible with the counit $\mathrm{Nor}^\mathrm{Nor}_*\mathbf{1} \rightarrow \mathbf{1}$.*

Proof. Here is the specific use of the absolute input. For a good pair, its Nori cohomology realizes as the algebraic relative de Rham cohomology and as relative Betti cohomology with their integral comparison. Morphisms of pairs, boundary maps, products, and Tate maps are included in this diagram. Apply the two realizations to the good-filtration complexes used to construct absolute derived Nori realization. They give the geometric realization complexes with the same maps: the augmentation to the geometric complex is the filtration spectral-sequence augmentation when each relative pair has cohomology in its designated degree.

Use the system of good filtrations and common refinements, together with finite affine Cech covers, not one independently chosen filtration for each variety. Its refinement maps and product maps are the complex-level construction in [11, §7] and [21, Chapters 4 and 7]. Applying the exact realization functors therefore gives a natural comparison of complexes, with the tensor coherences

supplied by that construction. Localization and Tate stabilization give (3.2), and Ind extends it from constructibles. This is the usual absolute mixed-realization factorization in External input 3.1. In particular, we do not need the more delicate comparison of separately chosen *relative* cellular filtrations discussed in Tubach’s Remark 4.10.

The de Rham–Betti comparison is natural on absolute Nori motives; apply that naturality to the counit, viewed as a morphism in $\mathrm{DN}(k)$. This proves the last assertion. No assertion that its matrix entries are algebraic is required. $\square$

3.2 The absolute augmentation and the relative module formula

Lemma 3.4 (Extension from an augmented algebra). *Let $\mathcal{A}$ and $\mathcal{B}$ be presentable stable symmetric monoidal categories, let $r : \mathcal{A} \rightarrow \mathcal{B}$ preserve colimits and tensor products, and let T be a commutative algebra of $\mathcal{A}$ with an augmentation $\alpha : r(T) \rightarrow \mathbf{1}_{\mathcal{B}}$. Then*

$$F(M) = \mathbf{1}_{\mathcal{B}} \otimes_{r(T)}^{\mathbf{L}} r(UM) \quad (3.3)$$

defines a colimit-preserving symmetric monoidal functor $\mathrm{Mod}_T(\mathcal{A}) \rightarrow \mathcal{B}$. It sends the free module $T \otimes Q$ to $r(Q)$. Among colimit-preserving r -linear functors with the specified identification on free modules, the functor and its compatible natural transformations are determined by r and the action α .

Proof. Applying r to the multiplication and action maps makes $r(UM)$ an $r(T)$ -module. Derived extension of scalars along α gives (3.3); associativity of relative tensor products gives its monoidal and colimit properties. For a free module the formula reduces to $\mathbf{1} \otimes_{r(T)} (r(T) \otimes r(Q)) \simeq r(Q)$.

For the determination assertion use the augmented simplicial free bar resolution of a T -module. Its terms are free modules on $T^{\otimes n} \otimes UM$, and its face maps are multiplication and the module action, with degeneracies induced by the unit. The underlying augmented simplicial object is split by the unit, so its geometric realization is M . A colimit-preserving r -linear functor with the given action on the unit sends this bar construction to the relative-tensor bar construction in (3.3). This proves the assertion for all modules and all module morphisms, not just for the objects that happen to be free. It also proves that a compatible comparison of r and α induces a coherent comparison of F . $\square$

By the lax symmetric monoidal right adjoint, $\mathcal{N}_k$ is a commutative algebra. Define an algebra map entirely over k by

$$\begin{aligned} \alpha_k : r_k(\mathcal{N}_k) &\simeq d_k(\mathrm{Nor}_k^* \mathrm{Nor}_{k,*} \mathbf{1}) \\ &\xrightarrow{d_k(\varepsilon)} d_k(\mathbf{1}) = k. \end{aligned} \quad (3.4)$$

Here ε is the adjunction counit. For $a_X : X \rightarrow \mathrm{Spec} k$ put $\alpha_X = a_X^* \alpha_k$ under $r_X(\mathcal{N}_X) \simeq a_X^* r_k(\mathcal{N}_k)$. The map α_k is a *de Rham augmentation*. It is not evaluation of a numerical period algebra at an alleged algebraic period point.

Theorem 3.5 (Algebraic realization and comparison interface). *For quasi-projective X/k , under (3.1) set*

$$R_{\mathrm{dR},X}(M) = \mathbf{1}_{\mathcal{D}_k(X)} \otimes_{r_X(\mathcal{N}_X)}^{\mathbf{L}} r_X(UM), \quad (3.5)$$

using α_X . On constructibles this is an algebraic realization over k with the following properties.

- (i) *It is symmetric monoidal, compatible with the six operations and their adjunction, projection, and base-change maps, and $R_{\mathrm{dR},X} \mathrm{Nor}_X^* = r_X$.*

- (ii) After $k \hookrightarrow \mathbb{C}$, regular Riemann–Hilbert identifies it naturally with the complexified Betti realization. Equivalently, its complex algebraic differential module is the underlying differential module of the established Nori-to-Hodge realization.
- (iii) It is t -exact for the perverse and holonomic hearts and is exact faithful on those hearts. It commutes with ordinary geometric nearby and vanishing cycles with their canonical and variation maps.
- (iv) Its absolute restriction is d_k , with the actual absolute comparison, not a newly chosen fiber functor.

The same statements hold after the k -linear coefficient extension of the rational Nori category.

Proof. Lemma 3.4 gives the functor, its tensor structure, and its formula on free modules. Since the augmentation is pulled back from $\mathrm{Spec} k$, associativity and pullback of relative tensor products supply the pullback comparison. On the absolute category the bar lemma identifies the formula with d_k : by construction (3.4) is exactly its action of $r_k(\mathcal{N}_k)$ on the unit.

For the complex comparison, write the established Betti functor on $\mathrm{DN}(X)$ in the module model. It is linear over the Betti realization of $\mathrm{DM}(X)$; its action on the unit is its own augmentation, pulled back from $\mathrm{Spec} k$. The absolute comparison and Lemma 3.3 identify this augmentation with $\alpha_k \otimes_k \mathbb{C}$ under the comparison for r_k . Pullback identifies the actions over X . The bar lemma now identifies the two functors on *all* modules and their morphisms. This proves (ii) as a natural enhanced comparison, rather than by assigning unrelated descent data to the Hodge realizations.

We give the six-operation verification explicitly. The compact module category is the thick closure of the free modules on compact geometric motives. Indeed, these modules are compact by the free-forgetful adjunction, since the forgetful functor preserves filtered colimits; they generate, since a module with no maps from their shifts has zero underlying object. For the explicit ordinary generators see [44, Corollary 4.19]. The free modules realize as constructible regular-holonomic objects by Lemma 3.2; finite cones and retracts show that the realization of every compact module is bounded constructible.

The exchange map for a direct image is the mate of the pullback comparison and the adjunction counit. On a free module $\mathrm{Nor}_Y^* Q$, the source uses $f_* \mathrm{Nor}_Y^* Q \simeq \mathrm{Nor}_X^* f_* Q$ from External input 3.1, and the exchange map becomes exactly $r_X f_* Q \rightarrow f_* r_Y Q$. It is the geometric exchange map, not merely an isomorphism between its source and target: the free action is multiplication followed by α_X , so naturality of the counit and the pulled-back augmentation identifies the two mates. This map is an equivalence by Lemma 3.2. The class of constructible objects on which an exchange map between exact functors is an equivalence is thick, so the claim follows for every compact module. The same argument applies to $f_!$ and the mate for $f^!$. Alternatively one obtains $f_!$ from open extension, proper direct image, and compactification. Duality is treated with the evaluation and dualizing object; internal Hom follows from constructible duality and tensor product. Thus all six comparisons exist and are isomorphisms on the category used here.

Their coherences are those of the module tensor construction and adjunction mates. In particular, a composition of pullbacks is identified by the associativity of the relative tensor product, and taking mates respects compositions. Projection and base-change squares reduce on free modules to the geometric squares and are compatible with the bar construction. We do not deduce equality of derived morphisms merely from conservativity of Betti realization.

After extension to $\mathbb{C}$, (ii) makes every perverse cohomology outside degree zero vanish for a Nori heart object. Faithful flatness of $k \hookrightarrow \mathbb{C}$ reflects that vanishing, proving t -exactness. If a heart object has zero algebraic realization, its Betti realization is zero; faithful Betti realization makes

the object zero. An exact functor detecting zero objects is faithful by applying this observation to images of morphisms. This also identifies our heart functor with the canonical universal-abelian-hull factorization of $Q \mapsto {}^p H^0 r_X(Q)$: after comparison its kernel is the same as that of perverse Betti realization, and the universal factorization is unique.

For ordinary geometric cycles one can use the actual gluing diagrams. Unipotent cycles are stabilized kernels and cokernels formed from $j_!, j_*$ and the logarithm objects whose algebraic connections are the nilpotent logarithmic systems $d - N dt/t$. Their units, trace maps, and nilpotent endomorphisms are geometric over k . The functor just constructed preserves these diagrams and their maps. Finite Kummer pullbacks and their character projectors supply the other eigenvalues for geometric objects; over $k = \overline{\mathbb{Q}}$ those projectors are defined. This proves the ordinary nearby/vanishing-cycle assertion including the canonical and variation maps. It does not identify arbitrary *irregular* moderate cycles with topological cycles; that separate issue is treated in Proposition 5.4.

Finally extend coefficients, not the geometric base. For a finite coefficient field $E/\mathbb{Q}$ in k , first apply the rational realization with its E -action, then take the summand corresponding to the specified embedding $E \hookrightarrow k$ in $E \otimes_{\mathbb{Q}} k$. Separability makes that projector exact. Every object, morphism, and finite diagram in the k -linear scalar extension is defined over such a finite E and then idempotent completion; all comparisons above commute with these projectors. Passing to their filtered union gives the final assertion. This is distinct from the geometric extension $k \hookrightarrow \mathbb{C}$. $\square$

Remark 3.6 (Scope of the realization construction). The relative algebraic functor, its operation maps, and the comparison are provided by (3.4)–(3.5). The relative tensor product in the formula is genuinely derived. Neither its t -exactness nor its faithfulness is automatic for a general augmented algebra; those conclusions were proved by the actual complex comparison and faithful scalar extension. Likewise, agreement on the objects of a generating family alone would not suffice without agreement on the augmentation and the bar-resolution maps. The standard external inputs are now the ordinary Nori module presentation, the ordinary geometric differential-module formalism, and the absolute mixed-realization construction. No relative algebraic realization theorem or relative period theorem is inserted as an assumption in their place. The construction is deliberately on enhanced categories: a statement only about isolated cohomology groups would not define the augmentation and bar comparison. The relative construction is separate from the final period argument; its inputs are the ordinary geometric and absolute realization theorems specified above.

3.3 The affine-line quotient

Fix X and $\pi : X \times \mathbb{A}^1 \rightarrow X$. Let $\mathcal{A}_X = \text{MN}(X)$ and $\mathcal{C}_X = \pi^*[1]\mathcal{A}_X \subset \mathcal{A}_{X \times \mathbb{A}^1}$.

Lemma 3.7 (Derived affine-line full faithfulness). *The adjunction unit $\text{id} \rightarrow R\pi_*\pi^*$ is an isomorphism. In particular π^* is fully faithful on the derived categories, in all degrees.*

Proof. Under Betti realization the unit is the usual pullback–pushforward unit for the product with the contractible complex affine line, and is an isomorphism. The cone has zero derived Betti realization; conservativity makes the cone zero. Adjunction now gives

$$\text{Hom}(\pi^*A, \pi^*B[n]) = \text{Hom}(A, B[n])$$

for every n . This is an assertion about the actual adjunction unit, not merely an unspecified equivalence of objects. $\square$

Lemma 3.8 (Reflection of pulled-back heart objects). *If $A \in \mathcal{A}_{X \times \mathbb{A}^1}$ and $\text{rat}(A) \simeq \pi^* L[1]$ for a perverse sheaf L on X , then $A \simeq \pi^* B[1]$, where $B = {}^p H^{-1}(R\pi_* A)$.*

Proof. The Betti realization of $R\pi_* A$ is $L[1]$. Its other perverse cohomology groups vanish, and exact faithfulness reflects their vanishing. Thus $R\pi_* A \simeq B[1]$. The counit $\pi^* B[1] \rightarrow A$ realizes as an isomorphism. Its cone is zero by conservativity. $\square$

Proposition 3.9 (Serre closure and the derived essential image). *The subcategory $\mathcal{C}_X$ is Serre. The essential image of derived π^* consists precisely of the bounded complexes whose perverse cohomology belongs to $\mathcal{C}_X$. It is thick and stable under perverse truncations.*

Proof. Smooth pullback with connected fibers has subquotient-closed image on perverse sheaves, by BBD, Corollaire 4.2.6.2. Exact Betti realization and Lemma 3.8 give subquotient closure in the Nori heart.

For extension closure, an extension of $\pi^* B[1]$ by $\pi^* A[1]$ is represented by a connecting morphism $\pi^* B[1] \rightarrow \pi^* A[2]$. By Lemma 3.7 this is the pullback of a morphism $B \rightarrow A[1]$. Its extension triangle is a pullback triangle, and $\pi^*[1]$ is t -exact; hence the middle heart object belongs to $\mathcal{C}_X$. This proves the Serre property using the special affine-line adjunction, not a claim about arbitrary connected smooth fibers.

The pullback of a bounded complex has cohomology in $\mathcal{C}_X$, up to the fixed shift. Conversely, induct on the number of nonzero perverse cohomology objects of a complex K . A single-cohomology object is in the image. In a truncation triangle, the two shorter objects are in the image by induction. Derived full faithfulness lifts the connecting morphism, so its cone is again a pullback. This proves the essential-image description. It also gives truncation stability. A summand is in the image because its cohomology is a summand of the pulled-back cohomology, hence belongs to the Serre subcategory. Thus the image is thick. $\square$

Remark 3.10 (Two invalid general principles are not used). The pullback image for $\mathbb{G}_m \rightarrow \text{Spec } k$ is not extension-closed: the connection $d - \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} dx/x$ is a nonconstant extension of two trivial connections. Likewise a Serre inclusion need not induce a fully faithful bounded derived functor. The proof above uses derived affine-line full faithfulness before truncation induction. It agrees with the division of responsibilities in [45, Proposition 7.2 and §7.3].

Definition 3.11 (Relative exponential Nori heart). Put $\text{EM}_N(X) = \text{MN}(X \times \mathbb{A}^1)/\pi_X^*[1]\text{MN}(X)$. The quotient heart and its realizations are constructed below.

Theorem 3.12 (The quotient heart and faithful realization). *The Verdier quotient*

$$D^b(\mathcal{A}_{X \times \mathbb{A}^1})/\pi^* D^b(\mathcal{A}_X)$$

has the induced bounded t -structure with heart $\text{EM}_N(X) = \mathcal{A}_{X \times \mathbb{A}^1}/\mathcal{C}_X$. Exponentiation supplies the derived six operations and derived convolution. Hodge realization descends to an exact faithful functor on the quotient heart. Composing with nonzero exponential realization gives an exact faithful ρ_λ , which reflects the bounded t -structure.

Proof. By Proposition 3.9, the localization subcategory is stable under truncations. The localization theorem for t -structures identifies the heart with the indicated Serre quotient. The derived operations are those of exponentiation [19]; this statement does not claim that all six operations are exact on every heart.

The Hodge functor maps the constant Serre subcategory into the corresponding Hodge subcategory. It thus induces an exact quotient functor. If a quotient object has zero Hodge image, any

heart representative has Hodge realization pulled back from X , hence Betti realization pulled back from X . Lemma 3.8 puts the representative in $\mathcal{C}_X$. The induced exact functor therefore detects zero objects, and is faithful: apply detection to the image of any morphism it kills. This argument does not require an arbitrary quotient morphism to lift to a morphism between particular chosen representatives.

Virk's Theorems 7.5–7.6 give exactness and faithfulness of nonzero realization on the exponential Hodge heart. Composition gives the same for ρ_λ . Finally $\rho_\lambda({}^p H^i K) = H^i(\rho_\lambda K)$, so exact faithfulness detects all cohomology vanishings and reflects the bounded t -structure. $\square$

Remark 3.13 (Absolute exponential presentation). At $X = \operatorname{Spec} k$, the vanishing-global-cohomology presentation of the exponential quotient gives the potential-line model used in the absolute exponential literature [45, 18]. The comparison needed here is constructed below on representatives and their quotient morphisms; no equivalence between two entire absolute motivic categories is used as an additional input.

3.4 The algebraic exponential functor and the matching Betti fiber

Theorem 3.14 (Algebraic nonzero-scale realization). *For $\lambda \in k^\times$ and an ordinary representative $A \in D^b(\operatorname{MN}(S \times \mathbb{A}^1))$, define*

$$\rho_{\lambda,k}(\mathbf{Q}_S A) = p!(R_{\operatorname{dR}, S \times \mathbb{A}^1}(A) \otimes \mu^* \mathcal{L}_{\lambda,k}), \quad \mathcal{L}_{\lambda,k} = \mathcal{K}_{\lambda,k}[-1]. \quad (3.6)$$

Here p and μ are the two projections and $\mathcal{K}_{\lambda,k} = (\mathcal{O}_{\mathbb{A}_k^1}, d - \lambda dt)$ is the holonomic heart object. The formula is a well-defined algebraic functor on the exponential derived quotient. It is tensor-compatible, commutes with $f^, f_*, f_!, f^!$, and has duality*

$$\mathbb{D}\rho_{\lambda,k} = \rho_{-\lambda,k}\mathbb{D}.$$

It is t -exact and faithful on the exponential heart. Its complexification is Virk's realization of the Hodge realization, with its actual operation maps. We henceforth denote this algebraic functor by ρ_λ .

Proof. All terms in (3.6) are over k by Theorem 3.5. The polynomial operator $\partial_t - \lambda$ is bijective on $k[t]$: on a polynomial v of degree m its inverse is

$$-\sum_{j=0}^m \lambda^{-j-1} \partial_t^j v.$$

Thus the algebraic direct image of the exponential kernel to a point vanishes; by duality the compactly supported direct image vanishes too. Projection formula now shows that the displayed functor kills every p^* -object. It therefore factors through the Verdier quotient, including all its morphisms, rather than merely being independent of representatives at the level of object classes.

The identities

$$\operatorname{sum}^* \mathcal{L}_{\lambda,k} \simeq \mathcal{L}_{\lambda,k} \boxtimes \mathcal{L}_{\lambda,k}, \quad i_0^* \mathcal{L}_{\lambda,k} = k$$

are the addition and zero-section identities for the shifted exponential kernel. Projection formula and Kunneth give tensor compatibility and the unit, with their associativity constraints; see also Virk's Propositions 6.4 and 6.8. Proper-support composition and base change first give $f_!$ and f^* compatibility. The natural forget-support map $p_! \rightarrow p_*$ after the twist is an isomorphism on ordinary regular input. Indeed, after $k \hookrightarrow \mathbb{C}$ it is exactly Virk's forget-support isomorphism, proved

in his Appendix B. Algebraic base extension is faithfully flat and commutes with the transfer complexes, so it reflects this isomorphism. Duality changes the sign of the exponential kernel, giving the displayed scale-reversing formula and the remaining f_* , $f^!$ compatibilities. These are the same projection, support, and adjunction maps as after complexification.

By Theorem 3.5, the complexified ordinary input is the underlying module of its Hodge realization. Thus the complexification of (3.6) is Virk's actual formula, not an arbitrary isomorphic connection. Virk's Theorems 7.5–7.6 prove t -exactness and heart faithfulness there. Faithful flatness reflects cohomology vanishings over k , and the faithful Hodge quotient realization of Theorem 3.12 reflects zero objects. Exactness then detects images of morphisms and proves faithfulness of $\rho_{\lambda,k}$. $\square$

Proposition 3.15 (Comparison with the chosen rapid-decay fiber). *Choose the direction used in (2.3) consistently with $e^{-\lambda t}$, and let $s_0 \in S(k)$ be a lisse point. The algebraic fiber $\omega_v(N) = \rho_\lambda(N)_{s_0}$ and the chosen Betti fiber have a natural tensor comparison*

$$c_N : \omega_{B,s_0}(N) \otimes_k \mathbb{C} \xrightarrow{\sim} \omega_v(N) \otimes_k \mathbb{C}.$$

On a sufficiently small simply connected lisse analytic open B about s_0 , these comparisons extend uniquely to a horizontal tensor comparison with the analytified algebraic connection. On elementary exponential data this is the usual rapid-decay integral comparison.

Proof. Normalized restriction $i_{s_0}^*[-d]$ commutes with the realization just constructed. At a point its de Rham functor is the absolute exponential de Rham functor: the ordinary part is d_k by Theorem 3.5(iv), and the exponential part is its twisted algebraic direct-image formula. The directional nearby-fiber functor at infinity and the absolute de Rham–rapid-decay comparison therefore give c_N , functorially in the absolute exponential object. Equivalently, compute on a regular representative on the potential line: the de Rham–rapid-decay theorem and the directional nearby-fiber identification give the comparison of its two vector spaces. The maps are natural in that representative and kill the constant subcategory, so they descend to the quotient. This uses the period comparison of these realizations, not an unproved equivalence of two entire absolute motivic categories. These are the absolute comparison inputs of [18, 23]; the directional convention and tensor fiber are recorded in [42, Definition 3.5 and Theorem 3.6]. Cohomological variance is used on both sides; a homology pairing is converted to this comparison by its duality, not by identifying a homology space with itself as a cohomology space.

Parallel transport of a flat connection extends each c_N uniquely over B . For a morphism $N \rightarrow M$, the two resulting maps of connections are horizontal and agree at s_0 , hence agree on B . The same argument for the tensor, symmetry, unit, and evaluation maps proves the tensor coherences. It applies to all subquotients because the absolute comparisons are already natural on the absolute exponential category and the algebraic realization is exact. Thus we are not inferring naturality from objectwise choices of a period matrix.

For elementary data, rapid-decay integrals are horizontal solutions of the twisted Gauss–Manin connection and their values at s_0 are the absolute comparison just used [24, 23]. Uniqueness of horizontal continuation makes the two comparisons equal on B . In particular the later identity $P = Yc$ uses this specific comparison, with no algebraicity assertion for the entries of c . $\square$

3.5 Consequences of Virk's realization theory

In this subsection $E(X)$ denotes the exponential Hodge derived quotient. The following consequences also apply to our compatible realizations where specified.

Proposition 3.16 (Derived conservativity and reflection of the t -structure). *Let $K \in E(X)$ and $\lambda \in \mathbb{C}^\times$. Then*

$$K \in E(X)^{\leq 0} \iff \mathrm{real}_\lambda(K) \in D_h^b(X)^{\leq 0},$$

and similarly for ≥ 0 . Thus real_λ is conservative on $E(X)$.

Proof. Virk proves t -exactness and faithfulness on the heart [45, Theorems 7.5–7.6]. Therefore

$${}^p H^i \mathrm{real}_\lambda(K) \simeq \mathrm{real}_\lambda({}^p H^i K).$$

If the left side vanishes, faithfulness on the heart gives ${}^p H^i K = 0$. Applying this degree by degree proves the assertions. $\square$

Proposition 3.17 (Exactness of the exponential Fourier transform). *Let $V \rightarrow S$ be a vector bundle. Virk's exponential Fourier transform*

$$\mathrm{FT}_V : E(V) \longrightarrow E(V^\vee)$$

is t -exact and induces an exact equivalence of exponential hearts.

Proof. Virk proves Fourier inversion and compatibility of FT_V with the classical D -module Fourier transform under real_1 [45, Theorems 5.3 and 6.12]. With Virk's shifts, the classical Fourier transform is t -exact. Hence the realization of $\mathrm{FT}_V(K)$ lies in the heart whenever K does. By Proposition 3.16, $\mathrm{FT}_V(K)$ lies in the exponential heart. Fourier inversion proves that the induced exact functor on hearts is an equivalence. $\square$

4 Fixed parts and extension from dense opens

Definition 4.1 (Lisse and admissible categories). An exponential heart object is λ -lisse when ρ_λ is an algebraic vector bundle with integrable connection. A λ -admissible subcategory is a strictly full rigid tensor subcategory of this lisse heart, with tensor product $\otimes_{\mathrm{lis}}$, closed under subquotients. Tannakian arguments always take place in such a category.

All tensor operations in this section are normalized as in (2.2). Call $N \in \mathrm{EM}_N(S)$ lisse when $\rho_\lambda(N)$ is an algebraic vector bundle with integrable connection. Exactness, faithfulness, and the usual closure properties of algebraic connections make this a Serre subcategory. Work in its dualizable rigid envelope when using Tannakian language.

Lemma 4.2 (Lowest direct image and independent horizontal sections). *For a connection M on a smooth connected d -fold,*

$$H^j(a_+ M) = 0 \quad (j < -d), \quad H^{-d}(a_+ M) = H_{\mathrm{dR}}^0(S, M) = \Gamma(S, M)^\vee.$$

Evaluation $\mathcal{O}_S \otimes_k \Gamma(S, M)^\vee \rightarrow M$ is injective and its image is the maximal trivial algebraic subconnection.

Proof. Algebraic de Rham hypercohomology has no negative ordinary degree; the perverse shift is $[d]$. Its lowest group is the kernel of the degree-zero connection operator on global sections. This gives the displayed formula also for a nonaffine base.

Horizontal sections independent over k are independent over $K = k(S)$. Indeed, in a shortest nontrivial rational relation normalize one coefficient to one and differentiate. Minimality forces

every remaining coefficient to be a common differential constant, hence to lie in k . This contradicts k -independence. Locally analytically, a horizontal section vanishing at one point vanishes identically; equivalently a fundamental solution matrix makes independent horizontal sections a constant full-rank matrix. Evaluation is therefore a bundle injection. Every trivial subconnection has a basis of global algebraic horizontal sections, proving maximality. $\square$

Theorem 4.3 (Algebraic exponential fixed part). *Let N be lisse and set*

$$C_N = {}^pH^{-d}(a_*N), \quad N^{\text{fix}} = \text{im}(a^*C_N[d] \rightarrow N).$$

*Then $\rho_\lambda(N^{\text{fix}})$ is the maximal trivial algebraic subconnection of $\rho_\lambda(N)$. An object has trivial connection realization if and only if it is constant, namely of the form $a^*C[d]$. The object N^{fix} is the maximal constant subobject of N .*

Proof. Operation compatibility and Lemma 4.2 make the perverse cohomology of a_*N vanish below $-d$ after realization. Reflection gives that vanishing in the motivic category. The lowest truncation map and adjunction therefore define the displayed morphism. It realizes as horizontal evaluation. Exact realization takes its image to the maximal trivial subconnection.

If Q realizes a trivial connection, its own evaluation map $a^*C_Q[d] \rightarrow Q$ realizes an isomorphism, and hence is an isomorphism. Conversely a constant object has constant connection realization. Apply this to N^{fix} . A morphism $a^*C[d] \rightarrow N$ corresponds by adjunction to $C[d] \rightarrow a_*N$. The lowest-cohomology truncation and the t -structure factor this map through $C_N[d]$. Thus every constant subobject is contained in N^{fix} . $\square$

Corollary 4.4 (Constants and their fiber functors). *Constant lisse objects are closed under subquotients, finite direct sums, normalized tensor products, and tensor duals. The functor $a^*[d]$ is fully faithful on the absolute heart. On the constant subcategory, the generic de Rham fiber is scalar extension from a k -valued absolute fiber functor, and its actual comparison matrix is independent of s .*

Proof. Subquotients of a trivial connection are trivial, so use Theorem 4.3. The other closure properties follow in the same way or from normalized pullback. For full faithfulness, adjunction computes $\text{Hom}(a^*C[d], a^*D[d])$ as $\text{Hom}(C, a_*a^*D)$. By the projection formula, $a_*a^*D = D \otimes a_*\mathbf{1}_S$. Connectedness and the degree-zero direct image give $H^0(a_*\mathbf{1}_S) = \mathbf{1}$ and no negative cohomology. Maps from C see just that degree-zero term, by truncation orthogonality. This gives $\text{Hom}(C, D)$. Pullback and comparison for the absolute object now give the fiber-functor statement. $\square$

Remark 4.5 (Algebraic, not merely analytic, invariants). On $\mathbb{A}_s^1$, $(\mathcal{O}, d - ds)$ has the analytic horizontal section e^s but no nonzero algebraic horizontal section: $p' = p$ has no nonzero polynomial solution. The fixed-part theorem takes algebraic horizontals. Finite Betti monodromy alone does not make an algebraic connection constant.

Let $j : U \hookrightarrow S$ be a dense open immersion. Intermediate extension in the exponential heart is the image of ${}^pH^0j_! \rightarrow {}^pH^0j_*$; its formation commutes with exact realization.

Lemma 4.6 (Lisse intermediate extension). *A lisse exponential object N is canonically $j_{!*}j^*N$. Restriction between the lisse subcategories is fully faithful. If $Q_U \in \text{EM}_N(U)$ realizes $L|_U$ for a connection L lisse on S , then $Q = j_{!*}Q_U$ is lisse and realizes L .*

Proof. A lisse connection on S has no nonzero holonomic subobject or quotient supported in $S \setminus U$, and is the minimal extension of its restriction. Exact realization reflects the absence of such boundary objects. The natural adjunction morphisms therefore identify N with $j_{!*}j^*N$.

A morphism on U induces a morphism of intermediate extensions. The canonical identifications just proved extend it to S ; a morphism with zero restriction has boundary-supported image, hence is zero. This proves full faithfulness. Finally $\rho_\lambda(j_{!*}Q_U) = j_{!*}(L|_U) = L$. Reflection places Q in the lisse heart. $\square$

Proposition 4.7 (Extending invertible lifts). *In the last assertion of Lemma 4.6, suppose Q_U is invertible. Then Q is invertible on S .*

Proof. Extend Q_U^{-1} by intermediate extension to Q' . Its realization is L^{-1} , so Q' is lisse. The products $Q \otimes_{\text{lis}} Q'$ and $Q' \otimes_{\text{lis}} Q$ are lisse and restrict to the unit on U . Applying Lemma 4.6 to these products and to the unit identifies both products with $1_{S, \text{lis}}$. These isomorphisms extend the inverse isomorphisms on U and satisfy their coherence identities by faithful restriction. $\square$

Remark 4.8 (What this does and does not say about opens). Every generic differential subquotient of a fixed lisse connection has a unique lisse model on S : take the saturated differential subobject in the holonomic category, then its quotient. Coherent modules with integrable connection on a smooth characteristic-zero variety are locally free, so no interior singularity is introduced. Morphisms extend by the same minimal-extension property. The preceding proposition allows an individually constructed motivic rank-one lift to be returned to that fixed base. It does not assert that the intersection of infinitely many dense opens is open, or that finite tensor generation means finitely many tensor constructions. The corresponding lisse-extension and full-faithfulness arguments apply to finite diagrams in the generic motivic category.

5 Fourier fibers and ordinary geometric constituents

For this section ordinary geometric input means a realization of an ordinary Nori object, before exponentiation. We will verify the smooth-projective realization of its relevant simple local-system constituents in Lemma 5.7 and Theorem 5.9. The comparison of the broader Nori and Ayoub geometric-origin categories in [27, Corollary 5.15 and Proposition 5.17] is background, not an input to that verification. No irregular Riemann–Hilbert correspondence is being used to algebraize a connection.

External input 5.1 (The Fourier comparison used here). For regular geometric input on $S \times \mathbb{A}_t^1$, use the normalized partial Fourier–Laplace transform. The algebraic argument in Lemma 5.2 gives a connection on $S \times \mathbb{G}_m$ after shrinking S . For a pure polarizable minimal extension $\mathcal{T}$ on $S \times \mathbb{P}_t^1$, Sabbah’s Proposition 5.8, Corollary 5.20, and sequences (5.21)–(5.22) compare the Fourier cycles at zero with ordinary cycles at $t = \infty$ [37]. Mixed input is reduced using its strict weight filtration and exact transform/cycle functors.

Lemma 5.2 (Finite graph reduction and nonzero-scale lissity). *For regular geometric perverse input K_* on $S \times \mathbb{A}_t^1$, after shrinking S there is a finite étale cover on which all dominating singular multisections split into pairwise disjoint graphs $t = u_i(s)$. The localized partial transform is a connection on $S \times \mathbb{G}_m$.*

Proof. Take a finite Whitney stratification adapted to the regular input. Delete the closures of the images of strata that do not dominate S . Each remaining proper stratum has dimension d : it

dominates a d -dimensional base and lies properly in a $(d+1)$ -fold. Its closure is generically finite over S . After taking the closures in $S \times \mathbb{P}^1$ and shrinking, these multisections miss infinity and are finite etale. A common finite Galois refinement splits them into pairwise disjoint sections. We use S for this cover until the final descent. For the underlying algebraic regular holonomic module M , the characteristic variety at finite t is contained in the zero section and the conormals

$$T_{\Gamma_i}^*(S \times \mathbb{A}_t^1) = \{a(dt - du_i) : a \in \mathbb{A}^1\}, \quad \Gamma_i = \{t = u_i(s)\}.$$

Since each u_i is regular, its graph is closed in $S \times \mathbb{P}^1$ and disjoint from $D_\infty = S \times \{\infty\}$. Thus the meromorphic extension $\overline{M} = j_+ M$, where $j : S \times \mathbb{A}^1 \hookrightarrow S \times \mathbb{P}^1$, is a meromorphic connection near D_∞ . It is regular there: algebraic regular holonomicity is preserved by this open direct image [20, Theorem 6.1.5(ii)]. Regularity at infinity is therefore a consequence of the input, not an additional hypothesis.

Set $Y = S \times \mathbb{G}_m$, with scale coordinate λ , and $Z = Y \times \mathbb{P}_t^1$, with proper projection $q : Z \rightarrow Y$. Pull back $\overline{M}$ as a differential module and twist by the Fourier kernel with connection $d - d(\lambda t)$. Denote the resulting meromorphic module by $\mathcal{N}$. Its direct image $q_+ \mathcal{N}$ computes the localized normalized partial transform. At finite t , twisting by a nonsingular connection leaves the characteristic variety unchanged, so its nonzero covectors are among $a(dt - du_i)$, with no $d\lambda$ -component. None annihilates the t -fiber direction.

Near infinity put $x = t^{-1}$. By the algebraic logarithmic-lattice criterion for regular connections [20, Theorem 5.3.7], choose a coherent lattice L in $\overline{M}$, with $L[x^{-1}] = \overline{M}$, stable under $x\partial_x$ and the parameter derivations ∂_{s_j} . Pull L back independently of λ . For the formal kernel generator e the sign convention above gives

$$\partial_\lambda e = -x^{-1}e, \quad \partial_x e = \lambda x^{-2}e.$$

For local generators m_a of L one has

$$\partial_\lambda^n(m_a e) = (-1)^n x^{-n} m_a e.$$

Consequently Le generates $\mathcal{N}$ as a $\mathcal{D}_Z$ -module. The filtration $F_p \mathcal{N} = \mathcal{D}_{Z, \leq p}(Le)$ for $p \geq 0$, and $F_p \mathcal{N} = 0$ for $p < 0$, is therefore exhaustive and good. The operators

$$x\partial_\lambda, \quad x\partial_x + \lambda\partial_\lambda, \quad \partial_{s_j}$$

preserve Le . They preserve every $F_p \mathcal{N}$: if P is one of these operators and Q has order at most p , then $P(Qv) = Q(Pv) + [P, Q]v$, with $[P, Q]$ of order at most p . Their degree-one symbols thus annihilate $\text{gr}_F \mathcal{N}$, and near infinity

$$\text{Char}(\mathcal{N}) \subseteq \{x\xi_\lambda = 0, \quad x\xi_x + \lambda\xi_\lambda = 0, \quad \xi_{s_j} = 0 \text{ for all } j\}. \quad (5.1)$$

For $x \neq 0$ this is the zero section. For $x = 0$, invertibility of λ forces $\xi_\lambda = 0$, leaving only the conormal to infinity. No nonzero covector in this conormal annihilates the t -fiber direction.

The proper direct-image characteristic estimate [20, Theorem 2.5.1 and Remark 2.5.2] now places the characteristic variety of every cohomology module of $q_+ \mathcal{N}$ in the zero section. Such a coherent differential module is an $\mathcal{O}_Y$ -coherent, hence locally free, integrable connection [20, Proposition 2.2.5]. Concentration in degree zero is a separate fact: the normalized partial Fourier transform is locally induced by the relative Weyl-algebra Fourier isomorphism and is exact; localization is exact as well. This proves the asserted lissity over k , which descends along the finite etale cover.

For comparison, the corresponding analytic result is [16, Theorem 1.11(2) and Corollary 1.12]. Its noncharacteristic condition follows from the displayed graph conormals: a covector pulled back from S has zero dt -component, so it intersects them only in zero. This reference is not needed for the algebraic characteristic-variety argument above. $\square$

Lemma 5.3 (Fixed fiber and topological nearby local system). *After the shrinking of Lemma 5.2, a fixed nonzero Fourier fiber and the topological nearby-cycle local system at $\lambda = 0$ have isomorphic underlying local systems on S , after forgetting the commuting nearby monodromy.*

Proof. First identify the algebraic fixed fiber. Write $\mathcal{F}_*$ for the localized transform, and let $i_{\lambda_0} : S \hookrightarrow S \times \mathbb{G}_m$ be the slice at $\lambda_0 \in k^\times$. For $K_* = R_{\text{dR}}(A)$ the identification is

$$i_{\lambda_0}^{*, \text{norm}} \mathcal{F}_* \simeq \rho_{\lambda_0}(\mathcal{Q}_S A)|_S. \quad (5.2)$$

Here normalized restriction means $i_{\lambda_0}^*[-1]$ in the perverse six-operation convention, or ordinary restriction of the underlying connection. Indeed, the characteristic bounds in Lemma 5.2 have no $d\lambda$ -component, so the slice is noncharacteristic both before and after direct image. Apply differential-module base change and noncharacteristic restriction [20, Theorems 1.7.3 and 2.4.6], then the forget-supports comparison in Theorem 3.14. The restricted kernel is $\mathcal{K}_{\lambda_0}$; the normalization gives $\mathcal{L}_{\lambda_0} = \mathcal{K}_{\lambda_0}[-1]$, exactly as in (3.6).

The local system on $S^{\text{an}} \times \mathbb{C}^\times$ is a representation of $\pi_1(S) \times \mathbb{Z}$. Restriction to a fiber forgets the second operator. Topological nearby cycles have the same representation of $\pi_1(S)$ and retain the second operator as nearby monodromy. Choose a path from the fixed scale to a punctured neighborhood of zero. Changing its winding changes the isomorphism by a power of the commuting operator, not the underlying isomorphism class or its constituents. $\square$

Proposition 5.4 (The regular-specialization bridge). *For the partial Fourier transform of the regular geometric input used in External input 5.1, moderate algebraic nearby cycles at $\lambda = 0$ realize the topological nearby cycles in Lemma 5.3. Consequently those moderate cycles and a fixed nonzero Fourier fiber have the same underlying S -local-system constituents after forgetting the nearby-monodromy operator. This assertion uses regular specialization along $\lambda = 0$, not just lissity on $S \times \mathbb{G}_m$.*

Proof. Work with the pushed-forward partial transform $\mathcal{F}$ on $S \times \mathbb{A}_\lambda^1$, before localizing in λ . Its analytification is regular at every finite Fourier scale by [16, Theorem 1.11(1)]. The hypotheses apply locally on S^{an} : on a sufficiently small relatively compact neighborhood the finitely many u_i are bounded, so one disc contains all the singular graphs. The regular meromorphic extension at $t = \infty$ was established in Lemma 5.2. No global boundedness of the graphs or purity of the input is required.

By Lemma 5.2, $\mathcal{F}|_{\lambda \neq 0}$ is a connection. Thus $\mathcal{F}(*\{\lambda = 0\})^{\text{an}}$ is a regular-singular meromorphic connection along the smooth divisor $\lambda = 0$. The classical logarithmic comparison now applies; see [15, II, Proposition 5.4, Remark 5.5(ii), and §6]. In a local logarithmic model $d + B d\lambda/\lambda$, the residue summands, with exponents taken modulo $\mathbb{Z}$, compute the moderate nearby cycles, while $\exp(-2\pi i B)$ is the nearby monodromy of the horizontal local system. Integrability makes this identification compatible with the S -connection. The local comparisons are natural and glue, with the perverse shift $[-1]$ on topological nearby cycles. Localization at $\lambda = 0$ changes neither nearby-cycle object. This proves the comparison directly on the transform, for mixed as well as pure regular input. Apply Lemma 5.3 for the constituent assertion.

For the comparison with ordinary cycles at infinity used below, retain Sabbah's kernel notation $\mathcal{FT} = p^+ \mathcal{T}[*\infty] \otimes e^{-t\lambda/z}$. Its regular specializability at finite λ is [37, Proposition 4.1(ii)];

Proposition 4.1(i) concerns the different, possibly irregular, $t' = 0$ direction. The projective cycle comparisons remain those of Proposition 5.8 and Corollary 5.20. The bridge just proved does not require a de Rham comparison for an arbitrary irregular module before projective direct image. $\square$

Remark 5.5 (The lissity-only assertion would be false). The connection $\mathcal{E}^{1/\lambda}$ on a punctured disc has a trivial rank-one analytic horizontal local system, but its formal type is purely irregular at zero and its moderate nearby cycles vanish. Its topological nearby local system has rank one. It is therefore essential to insert Proposition 5.4 before using the topological product argument in the algebraic-cycle proof. This example does not satisfy the regular-singular conclusion for the intended Fourier transform.

Proposition 5.6 (Unipotent comparison through two triangles). *Let $t' = t^{-1}$ and let $q : S \times \mathbb{P}_t^1 \rightarrow S$ denote the proper projection after taking cycles in the Fourier coordinate. Set*

$$V = \psi_{\lambda,-1}(\mathcal{FT}), \quad W = \ker N_\lambda.$$

Every simple constituent of every perverse cohomology of q_+V occurs among the perverse-cohomology constituents of the proper direct images of

$$i_{\infty,+} \ker N_{t'}, \quad \mathcal{T}, \quad i_{\infty,+} \psi_{t',-1} \mathcal{T}. \quad (5.3)$$

All cycles here are perverse-normalized.

Proof. Before q_+ , Sabbah's (5.22) and the first sequence of (5.21) give

$$0 \longrightarrow i_{\infty,+} \ker N_{t'} \longrightarrow W \longrightarrow \mathcal{T} \longrightarrow 0, \quad (5.4)$$

$$0 \longrightarrow W \longrightarrow V \longrightarrow i_{\infty,+} \psi_{t',-1} \mathcal{T} \longrightarrow 0. \quad (5.5)$$

Those sequences are stated before the subsequent dimension-one specialization in Sabbah's argument. Apply derived proper direct image to obtain two distinguished triangles. For any triangle $A \rightarrow B \rightarrow C \rightarrow$, the segment ${}^p H^j A \rightarrow {}^p H^j B \rightarrow {}^p H^j C$ gives a short exact sequence whose middle object is ${}^p H^j B$, whose subobject is a quotient of ${}^p H^j A$, and whose quotient is a subobject of ${}^p H^j C$. Apply this observation first to (5.4), then to (5.5), for each j . Finite length gives exactly the asserted constituent containment. No exactness of q_+ on the heart, and no interchange of a constituent filtration with a derived functor, is asserted. $\square$

Lemma 5.7 (Generic smooth-projective realization of constituents). *Let S/k be smooth, connected, and quasi-projective, and let $K \in D^b(\mathrm{MN}_{\mathbb{Q}}(S))$. Write $B(K)$ for its complex Betti realization. On a dense open $U \subset S$, the cohomology sheaves of $B(K)|_U$ are local systems, and every simple constituent is a direct summand of $R^i \pi_* \mathbb{C}$ for some smooth projective morphism $\pi : Y \rightarrow U$. The family and degree may depend on the constituent. A finite collection of K has a common such open. The assertion also holds after finite coefficient extension and passage to idempotent summands.*

Proof. Tubach's compact generators for $\mathrm{DN}(S)$ are $f_{\sharp} \mathbb{Q}_Y(n)$, with $f : Y \rightarrow S$ smooth and $n \in \mathbb{Z}$ [44, Corollary 4.19]. A compact object therefore belongs to a finite thick construction from finitely many such generators. Smooth purity identifies their Betti realizations, up to shifts and constant Tate twists, with $Rf_! \mathbb{C}_Y$. This uses the established ordinary Betti realization, not the new algebraic augmentation in Theorem 3.5.

A finite affine cover and the localization triangles reduce to quasi-projective Y . Take a projective compactification over S , then resolve it and principalize the boundary without changing Y [31,

Theorems 35–36]. We obtain $Y \subset \bar{Y}$, with $\bar{Y}$ smooth, $\bar{f} : \bar{Y} \rightarrow S$ projective, and $D = \bar{Y} \setminus Y$ a simple-normal-crossings divisor. The localization triangle is

$$Rf_! \mathbb{C}_Y \longrightarrow R\bar{f}_* \mathbb{C}_{\bar{Y}} \longrightarrow R(\bar{f}|_D)_* \mathbb{C}_D \longrightarrow .$$

The finite alternating resolution of $\mathbb{C}_D$ by the constant sheaves on the closed intersections D_I expresses the last term through smooth varieties projective over S . Thus $B(K)$ belongs to the thick subcategory generated by a finite list $Rp_{a,*} \mathbb{C}_{Z_a}$, where Z_a is smooth and $p_a : Z_a \rightarrow S$ is projective. The maps p_a need not yet be smooth; it is the closed intersections, not the open boundary strata, that supply these projective maps.

Remove the images of nondominating components and, for the remaining maps, the images of their nonsmooth loci. Properness and characteristic-zero generic smoothness give a common dense open U over which all p_a are smooth projective. Their cohomology sheaves $R^i(p_a|_U)_* \mathbb{C}$ are semisimple local systems [12, Proposition 4.2.5 and Theorem 4.2.6]. Let Σ be their finite set of simple constituents. Local systems whose composition factors belong to Σ form a Serre subcategory. Complexes with cohomology in that subcategory form a thick subcategory: the long exact cohomology sequence of a triangle uses only subquotients and extensions, and retracts preserve the same property. It follows that all simple constituents of $B(K)|_U$ belong to Σ . Semisimplicity of the displayed smooth-projective systems makes each such constituent an actual direct summand of one of them.

Tate twists do not change the underlying complex local systems. For a finite coefficient field, forget its action, apply the rational argument, and take the complex summand for the specified embedding. Idempotent completion introduces no new simple constituents. Apply the construction to a finite list at once to obtain the common-open assertion. No splitting of an arbitrary mixed local system, and no motivic lift of an arbitrary complex projector, is asserted. $\square$

Lemma 5.8 (Torsion of smooth-projective rank-one constituents). *A rank-one complex local system which, on a dense open of a smooth connected variety, is a constituent of $R^i \pi_* \mathbb{C}$ for a smooth projective π has finite monodromy.*

Proof. Semisimplicity makes it a direct summand on that open. Deligne’s determinant theorem for sub-local systems of algebraic variations gives finite tensor order [12, Corollary 4.2.8(iii)(b)]; equivalently, apply Lam–Litt’s published Lemma 3.5, whose statement allows a smooth variety of any dimension [33]. Its geometric-origin hypothesis is now satisfied by an actual smooth projective family. The integral structure needed for torsion comes from $R^i \pi_* \mathbb{Z}$ modulo torsion, not merely from a rational structure on a geometric-origin category. A complex rank-one summand need not itself have a rank-one integral lattice. Finally, inclusion of a dense open in a smooth connected variety induces a surjection on fundamental groups, so finiteness holds on the original lisse base as well. $\square$

Theorem 5.9 (Nearby-cycle geometricity). *Every simple lisse constituent of the moderate nearby-cycle object at $\lambda = 0$ of a partial Fourier transform of regular geometric input is, on a suitable common dense open, a direct summand of $R^i \pi_* \mathbb{C}$ for a smooth projective family π . In particular it is of ordinary geometric origin in the smooth-proper sense.*

Proof. We identify ordinary Nori objects containing the Betti constituents needed in the Fourier comparison. For a rational ordinary Nori representative M on $S \times \mathbb{A}^1$, take its weight pieces $M_w = \mathrm{gr}_w^W M$ in the Nori category. Existence and strictness of this filtration are [26, Corollary 6.16 and Proposition 6.17]. The established exact Nori-to-Hodge realization preserves weights [44,

Theorem 0.1]. Put $\overline{M}_w = j_{!*}M_w$, with $j : S \times \mathbb{A}^1 \hookrightarrow S \times \mathbb{P}^1$. Intermediate extension is an ordinary Nori operation and preserves purity [26, §6.7, Proposition 6.20]. Its pure Hodge realization decomposes into polarizable minimal-extension pieces $\mathcal{T}$ by strict support [40, §1.2 and Theorem 1.3]. Their underlying Betti objects are direct summands of $B(\overline{M}_w)$. We need only these summands of the realized object; we do not lift arbitrary complex projectors to motives.

Let $q : S \times \mathbb{P}^1 \rightarrow S$. For each such $\mathcal{T}$, additivity of derived direct image and of perverse cohomology gives

$${}^pH^a Rq_* B(\mathcal{T}) \quad \text{as a direct summand of} \quad B({}^pH^a q_* \overline{M}_w).$$

Here $B(\mathcal{T})$ denotes its complex Betti perverse sheaf. The object on the right is the realization of an ordinary Nori object on S . No exactness of q_* on the perverse heart is used.

For ordinary cycles at infinity use the chart $x = t^{-1}$. The perverse-normalized unipotent nearby-cycle functor is defined on ordinary Nori motives and is compatible with Betti realization [26, Corollary 3.14 and §3.5]. It therefore sends the containing Nori objects to ordinary Nori objects on S containing the required unipotent Betti summands. The other eigenvalues cause no change of category: geometric quasi-unipotence makes each monodromy eigenvalue ζ a root of unity; see [40, (1.1.5) and (2.1.6)–(2.1.7)]. On the punctured x -chart tensor with the Kummer local system of monodromy ζ^{-1} and take unipotent cycles. This extracts the ζ -part. The Kummer system is an Artin summand of a finite étale pushforward after a finite cyclotomic coefficient extension, so these are again ordinary Nori operations. The eigenvalue here is the Betti monodromy eigenvalue; Sabbah's index -1 denotes its unipotent part. Every constituent of $\ker N_\nu$ is already a constituent of the unipotent nearby-cycle object. Since $q \circ i_\infty = \text{id}_S$, its infinity-supported direct-image term requires no passage of an arbitrary subobject through a nonexact direct-image functor.

Now use Proposition 5.4. For pure input the non-unipotent Fourier comparison uses precisely the ordinary cycles just considered; the unipotent comparison uses the three sources in Proposition 5.6. Their simple lisse constituents thus occur in the Betti realizations of finitely many ordinary Nori objects on S . For mixed input, exactness of normalized Fourier transformation and nearby cycles, together with the strict finite weight filtration, reduces constituent containment to these pure pieces. Finite coefficient extensions are handled by the rational containing objects and their character summands, as in Lemma 5.7.

All weights, relevant perverse degrees, and eigenvalues form finite lists. Apply Lemma 5.7 to their containing Nori objects and shrink once. If the graph-splitting cover of Lemma 5.2 has been retained, the same conclusion holds downstairs: finite étale direct image is exact on local systems and is an ordinary Nori operation, while the unit and trace composite on a local system is multiplication by the degree. Thus its constituents downstairs occur in the direct images of the containing objects, to which Lemma 5.7 applies on S . This proves the stated smooth-projective realization, not just closure in a broadly defined geometric-origin category. $\square$

Corollary 5.10 (Finite Betti monodromy of rank-one factors). *A rank-one differential subquotient, defined over k , of a fixed nonzero Fourier fiber of regular geometric input has finite Betti monodromy on a suitable dense smooth open.*

Proof. By Lemma 5.3, Proposition 5.4, and Theorem 5.9, each rank-one Betti constituent of the fixed fiber is, on a common dense open, a direct summand of the cohomology of a smooth projective family. The local system of a rank-one differential subquotient is such a constituent because horizontal realization is exact on connections. Apply Lemma 5.8. This proves finite Betti monodromy, not regular singularity or finite tensor order of the original algebraic connection; the latter distinctions remain essential in Theorem 6.3. $\square$

Remark 5.11 (The two-graph family). Over the base $s \neq 0$, for the family $y^2 = t(t - s)$, the local system extends at $t = \infty$, so ${}^p\psi_{t', -1}\mathcal{T}$ has rank one, $N_{t'} = 0$, and trivial s -monodromy. In ordinary topological degrees the cohomology in degrees 0 and 2 vanishes because there are no invariant or coinvariant vectors. Since the local monodromies at $0, s$ are nontrivial, the middle-extension and punctured-curve Euler characteristics agree, and $\chi(\mathbb{P}^1 \setminus \{0, s\}, \mathcal{L}) = 0$; hence $q_+\mathcal{T} = 0$. Sabbah's upstream exact sequence gives $\dim \ker N_\lambda = 1$, whereas the unipotent nearby-cycle object has rank two. Thus N_λ has a single Jordan block and the two rank-one Jordan–Hölder constituents are trivial, as in the confluent system $M(1/2, 1, \lambda s)$.

6 The arithmetic kernel and rank-one lifts

Proposition 6.1 (Picard 1-motive convention). *Let U be smooth and let $\bar{U}$ be a smooth proper compactification with normal-crossings boundary D . With the covariance conventions used in this paper,*

$$T_H(\mathrm{Pic}^+(U)) \simeq H^1(U, \mathbb{Z}(1)), \quad \mathrm{Pic}^+(U) = [\mathrm{Div}_D^0(\bar{U}) \longrightarrow \mathrm{Pic}^0(\bar{U})]. \quad (6.1)$$

A morphism $\mathbb{Q}(0) \rightarrow \mathrm{Pic}^+(U)_\mathbb{Q}$ is represented by a rational boundary divisor in the kernel of the map to $\mathrm{Pic}^0(\bar{U}) \otimes \mathbb{Q}$. After multiplication by an integer making the divisor principal, say $\mathrm{div}(u)$, its de Rham realization is the logarithmic class $[d \log u]$.

Proof. This is the cohomological Picard 1-motive and its Hodge and de Rham realizations as constructed by Barbieri–Viale and Srinivas; see [8, Sections 2–4] and [13, Section 10]. $\square$

External input 6.2 (Degree-one arithmetic realization). Use the cohomological Picard 1-motive

$$\mathrm{Pic}^+(U) = [\mathrm{Div}_D^0(\bar{U}) \rightarrow \mathrm{Pic}^0(\bar{U})]$$

for a smooth compactification with normal-crossings boundary. Its realization is $H^1(U)(1)$. A rational boundary divisor giving a morphism from the weight-zero lattice object realizes, when made principal, as its logarithmic de Rham class [8]. The de Rham–Betti realization of 1-motives over $\bar{\mathbb{Q}}$ is fully faithful with image closed under subquotients [25, Theorem 9.14]. The comparison with degree-one Nori motives is [3].

Theorem 6.3 (Arithmetic exponential kernel). *Let U/k be smooth and connected and L a rank-one algebraic integrable connection over k with finite Betti monodromy. On a dense affine open U° there are $f \in \mathcal{O}(U^\circ)$ and a finite-order algebraic rank-one connection A such that*

$$L|_{U^\circ} \simeq \mathcal{E}^f \otimes A.$$

Proof. The finite Betti character defines a finite étale cover and its character idempotent over k ; algebraically closed extension does not change finite étale covers in characteristic zero. Let A_0 be its finite Artin connection and replace L by $L_0 = L \otimes A_0^{-1}$. On a dense affine open trivialize its underlying line bundle and write $\nabla = d + \omega$. Integrability gives $d\omega = 0$, and trivial monodromy gives

$$\int_\gamma \omega \in 2\pi i \mathbb{Z} \quad (\gamma \in H_1(U^{\mathrm{an}}, \mathbb{Z})).$$

Define the integral Betti class η by division of these periods by $2\pi i$. Comparison sends $[\omega]$ to $2\pi i \eta$. If $\eta = 0$, comparison implies $[\omega] = 0$, and affine algebraic de Rham theory supplies $F \in \mathcal{O}(U)$ with $\omega = dF$. Then $L_0 = \mathcal{E}^{-F}$.

Otherwise the *specified* pair $([\omega], \eta)$ gives the morphism

$$\mathbb{Q}(-1) \longrightarrow H_{\text{dRB}}^1(U), \quad (6.2)$$

with our convention that this Tate period is $2\pi i$. Twisting gives

$$\mathbb{Q}(0) \longrightarrow H_{\text{dRB}}^1(U)(1). \quad (6.3)$$

By External input 6.2, it lifts to the specified morphism $\mathbb{Q}(0) \rightarrow \text{Pic}^+(U)_{\mathbb{Q}}$. This is a rational boundary divisor Δ whose Picard class is zero in $\text{Pic}^0(\bar{U}) \otimes \mathbb{Q}$.

Choose $N > 0$ clearing its denominators and killing its torsion Picard class. Then $N\Delta = \text{div}(u)$ for a function with no zeros or poles on U , so $u \in \mathcal{O}(U)^\times$. Compatibility with the specified morphism gives

$$N[\omega] = [d \log u] \quad \text{in } H_{\text{dR}}^1(U/k).$$

Since U is affine, closed one-forms modulo exact global forms compute this cohomology. Thus $N\omega - d \log u = dF$ for some regular F , and

$$L_0 \simeq \mathcal{E}^{-F/N} \otimes (\mathcal{O}_U, d + N^{-1}d \log u).$$

The N th power of the second factor has horizontal algebraic generator u^{-1} and is trivial. Tensor back by A_0 . This proves the result without a relative functional-period theorem. $\square$

Proposition 6.4 (Realizing the algebraic factors). *At scale $\lambda \in k^\times$, $\mathcal{E}^f \otimes A$ in Theorem 6.3 has an invertible exponential motivic lift on U° . If the connection extends lisse to a given smooth base, so does an invertible lift.*

Proof. Use the graph potential f/λ in Lemma 2.1. Its inverse is the graph with the negative potential. A finite-order connection is a rank-one summand of a finite étale pushforward; over k its character idempotent defines a rank-one Artin motive. Insert it on the zero graph and take its normalized tensor product with the exponential graph object. Both factors are invertible and realize the desired factors. The extension assertion is Proposition 4.7. $\square$

Let $\eta = \text{Spec } k(S)$ be the generic point. We write

$$\text{EM}_N(\eta) := \varinjlim_{U \subset S} \text{EM}_N(U),$$

where objects and morphisms are identified after restriction to a smaller nonempty smooth open. Restriction is exact and compatible with the normalized tensor product, so every finite diagram in this filtered 2-colimit has a common model and the generic realization is an exact faithful tensor functor to algebraic differential modules over $k(S)$.

Let $\mathcal{T}_{\text{KZ}}(\eta)$ denote the filtered union of all finitely generated rigid Tannakian subcategories of $\text{EM}_N(\eta)$ generated by dualizable restrictions of elementary relative Kontsevich–Zagier exponential data.

Let $\mathcal{D}$ denote the differential tensor category generated by the realizations.

Definition 6.5 (Rank-one saturation). The pair $(\mathcal{T}, \rho_\lambda)$ is rank-one saturated if for every invertible $L \in \mathcal{D}$ there is an invertible object $\mathcal{L}$ in the ambient relative exponential Nori category such that

$$\rho_\lambda(\mathcal{L}) \simeq L.$$

Theorem 6.6 (Rank-one saturation on a fixed lisse model). *Let $\mathcal{T}$ be a finitely generated rigid subquotient-closed lisse subcategory of the generic exponential Nori envelope, modeled on S . Suppose its objects have regular geometric pre-Fourier representatives, as supplied by the exponential quotient and the ordinary geometric coefficient system. Let $\mathcal{D}$ be the differential tensor category generated over K by their realizations and differential subquotients. Every invertible object of $\mathcal{D}$ has an ambient invertible exponential motivic lift. The lift can be modeled on S whenever the differential object is modeled lisse there.*

Proof. An invertible object $L \in \mathcal{D}$ is a differential subquotient of a finite tensor construction in realized generators. Choose its motivic counterpart B and a common open model. The quotient description is essentially surjective on the heart, so choose an ordinary geometric representative K_B for B . At nonzero scale the representative realizes as its partial Fourier fiber. Ordinary tensor constructions, convolution, and the six operations stay within regular geometric input before exponentiation; no claim of regularity of the exponential output is made.

Apply Corollary 5.10 to L after shrinking. It has finite Betti monodromy there. It is algebraic over k , so Theorem 6.3 applies. Proposition 6.4 gives the ambient invertible lift on a dense affine open, and Proposition 4.7 returns it to S if needed. This is an object-by-object argument on one fixed lisse model, not an intersection of infinitely many opens. Rank-one objects with transcendental connection coefficients are not fed into the arithmetic theorem; extension of differential constants is handled later in Section 8. $\square$

Theorem 6.7 (Generic rank-one saturation). *For every $\lambda \in k^\times$, every finitely generated rigid Tannakian subcategory of $\mathcal{T}_{\text{KZ}}(\eta)$ is rank-one saturated.*

Proof. Every object of the exponential heart has an ordinary perverse-Nori representative. Its Hodge realization is regular geometric input, and convolution and the ordinary geometric operations retain that property before the Fourier realization. A finite collection has a common smooth open model. Apply Theorem 6.6 to a tensor construction containing the given rank-one differential subquotient. The extension statement in Proposition 4.7 returns the lift to any fixed lisse model when needed. No infinite intersection of separately chosen opens is taken. $\square$

6.1 An optional split criterion

Definition 6.8 (Split Fourier-geometric criterion). Let $\mathcal{T}$ be lisse at a scale $\lambda \in k^\times$ and let $\mathcal{D}$ be its differential tensor category. For each $B \in \mathcal{D}$, require a dense open and a finite Galois cover $q : S' \rightarrow S^\circ$ with the following properties:

- (FG1) Every rank-one differential subquotient L of B has an actual isomorphism $q^*L \simeq \mathcal{E}^{g_\alpha} \otimes A$, where $g_\alpha \in \mathcal{O}(S')$ and A is a rank-one regular-singular subquotient of one of finitely many regular-singular systems R_α .
- (FG2) On a common dense open, each simple constituent of the local system of every R_α is a direct summand of $R^i\pi_*\mathbb{C}$ for some smooth projective π . By Lemma 5.7, this holds for subquotients of ordinary Nori realizations. The whole possibly mixed system R_α need not be such a summand.
- (FG3) The descent datum on the resulting exponential–Artin lift is effective in the relative exponential Nori category.

The actual comparison in (FG1) is part of this sufficient criterion. An equality only of boundary Stokes grades does not assert it.

Lemma 6.9 (Rank-one comparison in the split criterion). *Under (FG1), each rank-one subquotient has the displayed exponential times regular-singular decomposition on the cover.*

Proof. This is precisely the actual rank-one comparison required in (FG1). When constructing an example from Stokes data, this global comparison must be proved in addition to the choice of a formal grade. $\square$

Lemma 6.10 (Finite regular factor). *Under (FG2), the factor A in Lemma 6.9 has finite monodromy and is a finite-order regular-singular connection.*

Proof. The rank-one local system of A is a simple constituent of one of the systems R_α . Condition (FG2) and Lemma 5.8 give finite monodromy, first on the common open and then on its lisse base. Regular-singular Riemann–Hilbert identifies A with the algebraic Artin connection of this finite character, so A has finite tensor order. $\square$

Theorem 6.11 (Split Fourier-geometric saturation). *The split criterion implies generic rank-one saturation. On a fixed lisse model the invertible lifts extend over that model.*

Proof. Apply Lemmas 6.9 and 6.10, realize the factors by the potential g_α/λ and the Artin idempotent, and use (FG3) for descent. Then apply Proposition 4.7. $\square$

This is an optional sufficient condition, not a hypothesis of Theorem 6.7. The latter uses the full nonzero-Fourier-fiber and arithmetic argument instead of a formal Stokes splitting shortcut.

7 Compatible fibers and Galois exactness

Choose $s_0 \in S(k)$ where a tensor generator is regular and fix the algebraic de Rham evaluation fiber

$$\omega_v : \mathcal{D} \longrightarrow \mathbf{Vec}_k, \quad M \longmapsto M_{s_0}.$$

All objects of $\mathcal{D}$ have their unique lisse model on S , as explained after Proposition 4.7. The same functor composed with realization is a fiber functor on $\mathcal{T}$. Set

$$G_v = \mathrm{Aut}^\otimes(\omega_v \rho_\lambda|_{\mathcal{T}}), \quad H_v = \mathrm{Aut}^\otimes(\omega_v|_{\mathcal{D}}).$$

The differential category is generated under subquotients by the image of $\mathcal{T}$, so the induced morphism $H_v \rightarrow G_v$ is a closed immersion, by [14, Proposition 2.21(b)]. This is not a consequence of exact faithfulness alone; the generation statement is also used.

Proposition 7.1 (Motivic semi-invariant subspaces). *For $B \in \mathcal{T}$ and a character $\chi : H_v \rightarrow \mathbb{G}_m$, the χ -semi-invariant subspace of $\omega_v \rho_\lambda(B)$ is the fiber of a subobject of B in $\mathcal{T}$.*

Proof. The character defines an invertible differential object L_χ . By Theorem 6.6, choose an ambient invertible lift $\mathcal{L}_\chi$, on a common lisse base if necessary. Take

$$C_\chi = (B \otimes_{\mathrm{lis}} \mathcal{L}_\chi^{-1})^{\mathrm{fix}}.$$

By Theorem 4.3 its connection realization is the maximal trivial subconnection of $\rho_\lambda(B) \otimes L_\chi^{-1}$. Tensor its inclusion by $\mathcal{L}_\chi$ and take the image $W_\chi \subset B$. Its fiber is exactly

$$L_\chi \otimes (\rho_\lambda(B) \otimes L_\chi^{-1})^{H_v},$$

namely the desired semi-invariant subspace. Although the lift was ambient, W_χ belongs to $\mathcal{T}$ because that category is closed under subobjects of its objects. $\square$

Lemma 7.2 (Normality from semi-invariants). *Let $H \subset G$ be algebraic groups over an algebraically closed field of characteristic zero. If V^χ is G -stable for every G -representation V and every character χ of H , then H is normal in G .*

Proof. By the stabilizer-of-a-line theorem for linear algebraic groups, choose a G -representation V and a line ℓ with stabilizer H . The line has an H -character χ and lies in $W = V^\chi$. By hypothesis W is G -stable and H acts on it by scalars. For $g \in G$ and $h \in H$, the element $g^{-1}hg$ therefore preserves ℓ , so it lies in H . Apply the same argument to g^{-1} to get equality. This proves normality on geometric points, hence on the characteristic-zero algebraic groups. This is the semi-invariant form of André's criterion used in [2, 34]. $\square$

Theorem 7.3 (Exactness in algebraic evaluation coordinates). *Let $\mathcal{T}_0$ be the full constant subcategory of $\mathcal{T}$, and let $G_{0,v}$ be its group under the evaluation fiber. Then*

$$1 \longrightarrow H_v \longrightarrow G_v \longrightarrow G_{0,v} \longrightarrow 1$$

is exact.

Proof. Proposition 7.1 and Lemma 7.2 make H_v normal. By Corollary 4.4, $\mathcal{T}_0$ is a full tensor subcategory closed under subquotients. Its inclusion gives a faithfully flat quotient of G_v [14, Proposition 2.21(a)]. A $\mathcal{T}$ -object is trivial as an H_v -representation precisely when its connection is trivial. By Theorem 4.3 this is equivalent to being constant. The constant subcategory is also closed under tensor duals, since the dual of a trivial connection is trivial. Thus the representations of G_v/H_v are precisely $\mathcal{T}_0$, proving exactness. No period-map injectivity has been used. $\square$

Corollary 7.4 (Split Fourier-geometric exactness). *For a finite tensor-generated category satisfying the split criterion, the exact sequence of Theorem 7.3 holds, on a common lisse model.*

Proof. Use Theorem 6.11 for rank-one lifts and Theorem 7.3 for the exact sequence. $\square$

7.1 Changing to Betti coordinates

Let $\omega_B = \omega_{B,s_0}$ and let ω_{dR} be the generic K -valued de Rham fiber. Define over k

$$G_B = \text{Aut}^\otimes(\omega_B|_{\mathcal{T}}), \quad G_{0,B} = \text{Aut}^\otimes(\omega_B|_{\mathcal{T}_0}), \quad H_B = \ker(G_B \rightarrow G_{0,B}). \quad (7.1)$$

Let $c : \omega_B \otimes_k \mathbb{C} \xrightarrow{\sim} \omega_v \rho_\lambda \otimes_k \mathbb{C}$ be the actual comparison at s_0 . It is an isomorphism of fiber functors, not an assertion that its entries are algebraic.

Lemma 7.5 (The group change is conjugation). *The comparison c identifies $(G_B)_\mathbb{C}$ with $(G_v)_\mathbb{C}$, and*

$$(H_B)_\mathbb{C} = c^{-1}(H_v)_\mathbb{C}c.$$

Proof. An isomorphism of fiber functors conjugates their tensor automorphism functors. Its restriction to $\mathcal{T}_0$ conjugates the quotient groups as well. Kernels are preserved by this commutative diagram. Use Theorem 7.3 to identify the evaluation-coordinate kernel with H_v . This proves the formula before any analytic torsor comparison. $\square$

8 Normalized solutions and extension of constants

The next argument establishes the differential base-change statement needed for the period comparison. It prevents arbitrary complex differential subobjects from being passed unexamined to an arithmetic theorem over k .

Shrink around s_0 to choose algebraic étale coordinates $x_1, \dots, x_d$ and a frame of a tensor-generator connection. The corresponding derivations ∂_i extend uniquely from $k(x_1, \dots, x_d)$ to K and commute. They span the tangent derivations over K . Write the system as

$$\partial_i Y = A_i Y, \quad A_i \in M_n(K), \quad Y(s_0) = I. \quad (8.1)$$

Here $\nabla = d - \sum_i A_i dx_i$; integrability is $\partial_i A_j - \partial_j A_i - [A_i, A_j] = 0$.

Lemma 8.1 (Algebraic Taylor normalization). *The solution Y of (8.1) exists on a sufficiently small analytic polydisc B and has Taylor coefficients in k .*

Proof. In étale coordinates, the completed local ring at s_0 is $k[[x_1, \dots, x_d]]$, with the coordinates translated to zero. The A_i have coefficients there. Coefficient comparison in (8.1), starting from I , determines each Taylor coefficient using only coefficients already determined and division by positive integers. Integrability is precisely the consistency of the different recursions. Analytic existence for a nonsingular flat system gives a convergent solution; uniqueness identifies its Taylor series with this formal solution. Thus all its coefficients lie in k . $\square$

External input 8.2 (Picard–Vessiot facts for integrable systems). For finitely many commuting derivations and an algebraically closed common constant field, a field generated by a fundamental matrix with no new common constants is Picard–Vessiot; its fundamental-matrix algebra is differentially simple. Its spectrum is the differential comparison torsor, and its group is the tensor automorphism group of the horizontal fiber functor. Use classical Picard–Vessiot theory and its integrable-system form [35, 30, 10]. This is ordinary Picard–Vessiot theory with all base derivations, not parameterized Picard–Vessiot theory with some derivatives omitted.

Put

$$R_k = K[Y, \det(Y)^{-1}] \subset \mathcal{M}(B), \quad E = \mathbb{C}(S).$$

The fraction field of R_k has common constants k . Indeed, a horizontal meromorphic germ is complex constant; after writing the germ as a quotient of two formal series over k , a nonzero coefficient of the denominator shows that this constant belongs to k . Thus External input 8.2 applies to R_k , and its solution fiber is the evaluation fiber ω_v , since $Y(s_0) = I$. Its group is H_v .

Lemma 8.3 (Finite coefficient descent). *For $f_1, \dots, f_r \in k[[x_1, \dots, x_d]]$, the space of complex constant relations among the f_j is the complex scalar extension of the space of k -relations.*

Proof. Expand $f_j = \sum_\alpha a_{\alpha j} x^\alpha$. A relation is the kernel of the matrix $(a_{\alpha j})$ with r columns. Its possibly infinite family of rows spans a finite-dimensional subspace of k^r . Choose finitely many rows that are a basis of this row span. They define the same kernel, and kernels of finite matrices commute with field extension. This proves the claim without a field automorphism or a continuity assumption on one. $\square$

Proposition 8.4 (No extra relations after complex constant extension). *Multiplication of meromorphic functions gives an isomorphism*

$$R_k \otimes_K E \xrightarrow{\sim} E[Y, \det(Y)^{-1}] =: R_E. \quad (8.2)$$

The algebra R_E is Picard–Vessiot over E with group $(H_v)_\mathbb{C}$.

Proof. Surjectivity is immediate. To prove injectivity, take a finite expression in the kernel. Choose an affine neighborhood of s_0 defined over k . Every coefficient in E is a quotient of finite complex linear combinations of regular k -functions. Clear a common nonzero denominator of this type. The alleged relation becomes a finite expression

$$\sum_{j=1}^r c_j r_j = 0, \quad c_j \in \mathbb{C}, \quad r_j \in R_k.$$

Multiply once more by a nonzero element of K , if necessary, to remove the finitely many poles at s_0 of the r_j . Their expansions are now in $k[[x_1, \dots, x_d]]$ by Lemma 8.1. By Lemma 8.3, the vector (c_j) is a complex linear combination of k -relation vectors. Those relations already vanish in R_k . Therefore the tensor expression was zero. Reversing the invertible denominator multiplications proves injectivity.

The fraction field of R_E is a subfield of $\mathcal{M}(B)$ containing $\mathbb{C}$, so its common constants are exactly $\mathbb{C}$. It is generated by a fundamental matrix, hence is Picard–Vessiot by External input 8.2. For the group identification, the Picard–Vessiot torsor identity for R_k is

$$R_k \otimes_K R_k \simeq R_k \otimes_k k[H_v].$$

Tensor it with E and use (8.2). This is the torsor identity for R_E with group $(H_v)_{\mathbb{C}}$. The action sends Y to Yh . Equivalently, the matrix relation ideal and its tensor symmetries have just been shown to commute with this constant extension. $\square$

Remark 8.5 (Why this base-change argument is needed). An exact faithful functor on the original k -category does not by itself identify every later complex differential subobject. Here that step is proved using a normalized solution with algebraic Taylor coefficients. Actual period matrices need not have algebraic Taylor coefficients; they are introduced only after Proposition 8.4.

9 The torsor and the exact constant-field argument

We now give the comparison and descent argument. The result also has an abstract formulation: it applies to any finite-type neutral Tannakian system with the fiber functors and the exactness established in Section 7, and with the algebraic connection realization used in Section 8.

9.1 The actual constant-period base

Let $\omega_{\text{dR},0} : \mathcal{T}_0 \rightarrow \text{Vec}_k$ be the constant de Rham fiber of Corollary 4.4. Define the comparison torsors

$$\begin{aligned} X &= \text{Isom}^{\otimes}(\omega_B \otimes_k K, \omega_{\text{dR}}), \\ X_0 &= \text{Isom}^{\otimes}(\omega_B|_{\mathcal{T}_0}, \omega_{\text{dR},0}). \end{aligned}$$

Thus X is over K , whereas X_0 is over k . In a fiber product the base change $X_{0,K}$ is always understood explicitly. These are affine torsors by [14, Theorem 3.2].

Let $p_0 \in X_0(\mathbb{C})$ be the actual constant comparison point and put

$$\begin{aligned} Z_0 &= \overline{\{p_0\}}^{\text{Zar}}, & A_0 &= k[Z_0] = k[X_0]/\ker(\text{ev}_{p_0}), \\ F_0 &= \text{Frac}(A_0), & R_0 &= K \otimes_k A_0, & K_0 &= \text{Frac}(R_0) = F_0(S). \end{aligned} \tag{9.1}$$

Both A_0 and F_0 are identified with their actual embedded values in $\mathbb{C}$. This imposes their relations without asserting that the unknown constant-period relations are absent. All derivations kill F_0 .

Lemma 9.1 (Domains, embeddings, and base constants). *There are injective maps*

$$R_0 \hookrightarrow K_0 \hookrightarrow E = \mathbb{C}(S),$$

and the common constant field of K_0 is exactly F_0 .

Proof. Evaluation makes A_0 a domain. The smooth connected variety S is geometrically integral over the algebraically closed field k . After any field extension its function-field construction is therefore a domain. In particular $K \otimes_k \mathbb{C}$ is a domain whose fraction field is E . Flatness of K over k embeds R_0 in $K \otimes_k \mathbb{C}$, giving the claimed injections and identifying K_0 with $F_0(S)$.

The derivations ∂_i form a basis of the tangent derivations of K_0/F_0 . If they all kill $u \in K_0$, then $du = 0$ in Ω_{K_0/F_0}^1 . In characteristic zero this implies that u is algebraic over F_0 . Geometric integrality of S_{F_0} implies that F_0 is relatively algebraically closed in $F_0(S)$, so $u \in F_0$. The reverse inclusion is part of the definition of the derivations. $\square$

Set

$$X_{\text{rel}} = X \times_{X_{0,K}} (\text{Spec } K \times_k Z_0), \quad B_0 = \mathcal{O}(X_{\text{rel}}). \quad (9.2)$$

Proposition 9.2 (The correct group and its torsor). *The map $X_{\text{rel}} \rightarrow \text{Spec } R_0$ is a torsor under H_{B,R_0} . Its coordinate algebra B_0 is faithfully flat and torsion-free over R_0 .*

Proof. The normal subgroup H_B of (7.1) is defined over k . The map $X \rightarrow X_{0,K}$ is an $H_{B,K}$ -torsor: after a faithfully flat trivialization of X , it is the quotient map $G_B \rightarrow G_{0,B}$, whose kernel is H_B . Descend this assertion and base change along $\text{Spec } K \times_k Z_0 \rightarrow X_{0,K}$. An affine torsor under an affine algebraic group is faithfully flat. Since R_0 is a domain, flatness implies torsion-freeness. $\square$

There is a natural differential structure on B_0 . For a tensor-generator connection in the frame of (8.1), its universal comparison matrix U satisfies

$$\partial_i U = A_i U. \quad (9.3)$$

Naturality and tensor relations are stable under these operators because the de Rham morphisms and tensor structure are horizontal. Constant comparison coefficients are horizontal; hence restriction along Z_0 also respects the derivations. Thus actual analytic comparison defines a differential map $B_0 \rightarrow \mathcal{M}(B)$ without yet claiming it injective.

9.2 An explicit equivariant comparison over $\mathbb{C}(S)$

Use the tensor generator, algebraic frame, and normalized matrix Y of Section 8. Let P be its actual comparison matrix from Betti to de Rham coordinates, and let $c = P(s_0)$ be the matrix of the comparison in Lemma 7.5. Uniqueness of solutions gives

$$P = Yc. \quad (9.4)$$

The coefficients of c belong to $\mathbb{C}$ and need not belong to F_0 . This fact is allowed; those coefficients are not adjoined to the smaller solution field in the descent argument.

Lemma 9.3 (The analytic algebra and equivariance). *Evaluation gives a surjective E -algebra map*

$$B_0 \otimes_{R_0} E \longrightarrow E[P, \det(P)^{-1}] = R_E. \quad (9.5)$$

The induced closed immersion of schemes is equivariant for $H_{B,E}$, under the identification $(H_B)_{\mathbb{C}} = c^{-1}(H_v)_{\mathbb{C}}c$.

Proof. The actual comparison is natural and tensor-compatible. Its restriction to constant objects is p_0 , so it factors through X_{rel} . Because $\mathcal{T}$ has a tensor generator, its comparison algebra is generated by the entries of the generator's universal matrix and its inverse determinant, with its tensor relations. Evaluation therefore has precisely the image on the right of (9.5), and is surjective. By (9.4), $E[P, \det(P)^{-1}] = E[Y, \det(Y)^{-1}]$.

The group of the latter Picard–Vessiot algebra acts by $Y \mapsto Yh$, $h \in (H_v)_{\mathbb{C}}$. Consequently its action on P is

$$P = Yc \longmapsto Yhc = P(c^{-1}hc).$$

By Lemma 7.5, $c^{-1}hc$ ranges over $(H_B)_{\mathbb{C}}$. On the motivic comparison torsor, precomposition by a Betti tensor automorphism is exactly right multiplication $U \mapsto Ug$. The two formulas coincide. They are identities for the universal matrix coordinate functions, so they establish equivariance as group-scheme actions, not only on a selected set of analytic points. $\square$

Lemma 9.4 (Equivariant torsor maps). *An equivariant morphism between two nonempty torsors under the same affine group over a field is an isomorphism.*

Proof. Make a faithfully flat extension that trivializes both torsors and choose origins. An equivariant map between the two right regular group torsors is left translation by the image of the identity, hence is invertible. The inverse descends, since invertibility of a morphism is faithfully flat local on the base. $\square$

Theorem 9.5 (Injective relative period comparison). *The actual map $B_0 \rightarrow \mathcal{M}(B)$ is injective. Moreover*

$$B_0 \otimes_{R_0} \mathbb{C}(S) \simeq R_E \tag{9.6}$$

as differential algebras and equivariant torsor coordinate algebras.

Proof. By Proposition 8.4, the source scheme of the closed immersion in Lemma 9.3 is a Picard–Vessiot torsor under $H_{B,E}$, in the Betti coordinates just computed. By Proposition 9.2, its target is a torsor under the same group. Lemma 9.4 makes that immersion an isomorphism. This proves (9.6).

By Proposition 9.2 and Lemma 9.1, the actual period map is the composite of injections

$$B_0 \hookrightarrow B_0 \otimes_{R_0} K_0 \hookrightarrow B_0 \otimes_{R_0} E \xrightarrow{\sim} R_E \hookrightarrow \mathcal{M}(B).$$

The first injection is localization of a torsion-free module over a domain. The second is scalar extension along the field embedding $K_0 \hookrightarrow E$, hence is injective. The last is the concrete analytic Picard–Vessiot embedding. This proves injectivity without assuming anything about the constant field of the smaller period-generated solution field. $\square$

9.3 Descent of differential simplicity and constants

Put

$$R = B_0 \otimes_{R_0} K_0, \quad L = \text{Frac}(R).$$

These are domains by Theorem 9.5. In particular $R = K_0[P, \det(P)^{-1}]$ in the actual analytic embedding, not an algebra obtained by adjoining the separate entries of c .

Lemma 9.6 (Intersection in a faithfully embedded tensor product). *Let $F \subset L_1, L_2$ be fields. In $L_1 \otimes_F L_2$, equality $x \otimes 1 = 1 \otimes y$ with $x \in L_1$ and $y \in L_2$ implies $x = y \in F$. The same conclusion holds if the equality is checked in any field into which this tensor product injects.*

Proof. If $x \notin F$, the vectors $1, x$ are linearly independent over F . Choose an F -linear functional $\ell : L_1 \rightarrow F$ with $\ell(1) = 0$ and $\ell(x) = 1$, extending from their span to a basis. Applying $\ell \otimes \text{id}$ to the equality gives $1 = 0$, a contradiction. Thus $x \in F$, and the equality then implies $y = x$ by injectivity of the field into its scalar extension. The last assertion reduces to this one by the assumed injection. $\square$

Theorem 9.7 (Picard–Vessiot descent over the actual constant field). *The algebra R is differentially simple, its fraction field satisfies*

$$L^{\partial_1=\dots=\partial_d=0} = F_0, \quad (9.7)$$

and R is the Picard–Vessiot algebra for the realized tensor generator over $K_0 = F_0(S)$, with group $(H_B)_{F_0}$.

Proof. By Theorem 9.5, $R \otimes_{K_0} E \simeq R_E$ is differentially simple. If $I \subset R$ is a nonzero differential ideal, faithful flatness of the field extension makes $I \otimes E$ nonzero. Simplicity makes it all of R_E . Applying faithful flatness to $(R/I) \otimes E = 0$ gives $I = R$. Thus R is differentially simple.

To prove (9.7), localize the isomorphism $R \otimes_{K_0} E \simeq R_E$ at every nonzero element of R . Since all those elements map to nonzero elements of the domain R_E , this gives an injection

$$L \otimes_{K_0} E \hookrightarrow \text{Frac}(R_E) \hookrightarrow \mathcal{M}(B). \quad (9.8)$$

Let $b \in L$ be killed by every ∂_i . Its analytic image is a constant $\alpha \in \mathbb{C} \subset E$: on a connected polydisc a meromorphic function with all partial derivatives zero is constant. In the last field of (9.8) the two elements $b \otimes 1$ and $1 \otimes \alpha$ have equal image. The injection gives their equality in the tensor product. Lemma 9.6 implies $b \in K_0$. Its derivatives are zero there, so Lemma 9.1 gives $b \in F_0$. Conversely, F_0 is killed by all the derivations.

Finally R is generated over K_0 by the actual fundamental matrix P and its inverse determinant, it is differentially simple, and its fraction field has exactly the constants F_0 . These are the Picard–Vessiot properties over this possibly non-algebraically-closed constant field. The already established torsor coaction is defined over F_0 and identifies its group with $(H_B)_{F_0}$; equivalently this identification follows after the faithfully flat constant extension to $\mathbb{C}$ from Lemma 9.3 and descends. This completes the constant-field argument rather than presupposing it. $\square$

For $d = 0$, $S = \text{Spec } k$, the relative group is trivial and the argument reduces to the embedding $A_0 \hookrightarrow \mathbb{C}$; set $K_0 = L = F_0$. For $d > 0$, the proofs above use every base derivation. Injectivity on B implies injectivity on any common analytic continuation domain, so the same result holds on the universal-cover formulation of the period map. The passage from a finite tensor-generated category to the generic exponential envelope is given in Section 10.

Theorem 9.8 (Relative functional exponential-period theorem). *For a category satisfying the fixed-part and rank-one saturation properties proved above, the generic fiber of X_{rel} is the Picard–Vessiot torsor over $K_0 = F_0(S)$. The actual relative period map*

$$\text{per}_{S, \text{rel}}^{\text{exp}} : B_0 \longrightarrow \mathcal{M}(\tilde{S})$$

is injective.

Proof. The injection is Theorem 9.5. Differential simplicity, the exact field of constants, and the torsor identification over K_0 are Theorem 9.7. The preceding argument on a connected analytic open determines the continued map on $\tilde{S}$. $\square$

Corollary 9.9 (Functional relation interpretation). *Under rank-one saturation, every polynomial functional relation among the relative exponential periods of $\mathcal{T}$, with coefficients in the actual constant exponential-period field, is induced by a tensor relation in $\mathcal{T}$.*

Proof. Polynomial expressions in matrix coefficients are matrix coefficients of finite tensor constructions. Injectivity of the comparison algebra therefore identifies the kernel of functional evaluation with the tensor relations defining the comparison torsor. $\square$

Remark 9.10 (Categorical and elementary period algebras). Corollary 9.9 gives completeness in the motivic formal-period algebra. It does not by itself prove that this algebra has a minimal presentation solely by linearity, elementary algebraic changes of variables, Fubini, and twisted Stokes. The comparison between the categorical presentation and a chosen elementary Kontsevich–Zagier presentation is a separate theorem.

10 Relation with Kontsevich–Zagier exponential periods

Let $\mathcal{T}_{\text{KZ}}(\eta)$ be the generic relative exponential Nori Tannakian envelope introduced before Theorem 6.7. It is generated by the restrictions of elementary data

$$(X \rightarrow S, Y, f, n, i), \quad f : X \rightarrow \mathbb{A}^1 \text{ regular},$$

and their tensor operations and duals. Every finite packet has a common dense lisse model. The comparison theorems for exponential periods identify its matrix coefficients with cohomological rapid-decay period functions [9, 18].

Theorem 10.1 (Relative exponential Nori–Ayoub theorem). *For the canonical realization $\lambda = 1$, the generic relative formal exponential-period map of the filtered Kontsevich–Zagier exponential Tannakian envelope is injective after the algebraic relations among constant exponential periods have been imposed. The same statement holds for every $\lambda \in \overline{\mathbb{Q}}^\times$. Equivalently, every algebraic functional relation among a finite packet of relative exponential periods is induced, on a common dense lisse open, by a tensor relation in the relative exponential Nori category.*

Proof. Every finite packet is contained in a finitely generated rigid Tannakian subcategory of $\mathcal{T}_{\text{KZ}}(\eta)$. By Theorem 6.7, that subcategory is rank-one saturated at the chosen algebraic nonzero scale. Choose a common dense lisse model and apply Theorem 9.8. Taking the filtered colimit over finite packets proves the generic statement. Since analytic functions agreeing on a dense open agree wherever both sides are defined, this is the asserted functional relation theorem. $\square$

10.1 Arithmetic specialization

Theorem 10.1 does not solve the numerical exponential-period conjecture. It controls identities valid identically in the parameters. A relation created at an algebraic fiber requires an arithmetic lifting theorem. In the E -function setting this role is played by Beukers’ refinement of the Siegel–Shidlovskii theorem; in period-valued, Gamma, and more general irregular settings the corresponding specialization theorems remain independent problems. The cyclotomic Gamma calculation in Section 11.2 makes this separation explicit.

11 Applications and examples

11.1 Polynomial exponential integrals

For a finite packet of integrals

$$F_j(z) = \int_{\gamma_j} p_j(x) e^{-z q_j(x)} dx,$$

the Fourier support is finite and algebraic. Whenever the companion polynomial-line analysis verifies that a finite cover splits the critical-value support and that the regular coefficient systems have finite monodromy, the generic split Fourier-geometric theorem applies. The present result then gives the categorical completeness of functional relations; specialization at an algebraic value of z remains the separate arithmetic E -function step.

11.2 Cyclotomic Gamma packets and the Rohrlich–Lang boundary

The main theorem gives categorical completeness for these rapid-decay periods. For the standard one-parameter Gamma packet, the complete functional ideal can also be computed directly. The resulting Gamma-specific presentation is elementary and, to our knowledge, has not been recorded in this form.

Fix an integer $N \geq 2$ and distinct integers

$$1 \leq m_1 < \cdots < m_s < N.$$

Put

$$r_j = \frac{m_j}{N}, \quad \gamma_j = \Gamma(r_j), \quad C_\Gamma = \overline{\mathbb{Q}}(\gamma_1, \dots, \gamma_s) \subset \mathbb{C}.$$

For $\Re z > 0$, define

$$G_j(z) = N \int_0^\infty x^{m_j-1} e^{-zx^N} dx,$$

and continue the function to the universal cover $\widetilde{\mathbb{G}_m}$. For a multi-index $\mathbf{a} = (a_1, \dots, a_s) \in \mathbb{N}^s$, write

$$X^{\mathbf{a}} = \prod_{j=1}^s X_j^{a_j}, \quad \gamma^{\mathbf{a}} = \prod_{j=1}^s \gamma_j^{a_j}, \quad M(\mathbf{a}) = \sum_{j=1}^s a_j m_j.$$

Theorem 11.1 (Functional cyclotomic Gamma presentation). *One has*

$$G_j(z) = \gamma_j z^{-m_j/N}.$$

Let

$$\Phi : C_\Gamma(z)[X_1, \dots, X_s] \longrightarrow \mathcal{M}(\widetilde{\mathbb{G}_m}), \quad X_j \longmapsto G_j(z).$$

Then

$$\ker \Phi = \left(\gamma^{\mathbf{b}} X^{\mathbf{a}} - z^{-q} \gamma^{\mathbf{a}} X^{\mathbf{b}} : M(\mathbf{a}) - M(\mathbf{b}) = qN \right), \quad (11.1)$$

where $\mathbf{a}, \mathbf{b} \in \mathbb{N}^s$ and $q \in \mathbb{Z}$. Thus every polynomial functional relation in the packet is generated by the Kummer-weight binomials in (11.1).

Proof. For positive real z , the substitution $u = zx^N$ gives

$$G_j(z) = z^{-m_j/N} \int_0^\infty u^{m_j/N-1} e^{-u} du = \gamma_j z^{-m_j/N}.$$

Both sides are holomorphic on the right half-plane, so the identity holds there and then on $\widetilde{\mathbb{G}_m}$ by analytic continuation.

Choose the global function

$$t = z^{-1/N}$$

on $\widetilde{\mathbb{G}_m}$. Then $t^N = z^{-1}$ and

$$G_j(z) = \gamma_j t^{m_j}.$$

Since $\mu_N \subset \overline{\mathbb{Q}} \subset C_\Gamma$, the cyclic action $t \mapsto \zeta t$ has fixed field $C_\Gamma(t^N) = C_\Gamma(z)$. Hence

$$[C_\Gamma(t) : C_\Gamma(z)] = N, \quad 1, t, \dots, t^{N-1}$$

is a basis over $C_\Gamma(z)$.

Let I denote the ideal on the right-hand side of (11.1). Every displayed generator maps to zero, since $M(\mathbf{a}) - M(\mathbf{b}) = qN$ implies

$$t^{M(\mathbf{a})} = z^{-q} t^{M(\mathbf{b})}.$$

Thus $I \subseteq \ker \Phi$.

Conversely, let

$$P = \sum_{\mathbf{a}} c_{\mathbf{a}}(z) X^{\mathbf{a}} \in \ker \Phi.$$

Write uniquely

$$M(\mathbf{a}) = q_{\mathbf{a}}N + \rho_{\mathbf{a}}, \quad 0 \leq \rho_{\mathbf{a}} < N.$$

Then

$$0 = \Phi(P) = \sum_{\rho=0}^{N-1} t^\rho \left(\sum_{\rho_{\mathbf{a}}=\rho} c_{\mathbf{a}}(z) \gamma^{\mathbf{a}} z^{-q_{\mathbf{a}}} \right).$$

Linear independence of $1, t, \dots, t^{N-1}$ gives, for every ρ ,

$$\sum_{\rho_{\mathbf{a}}=\rho} c_{\mathbf{a}}(z) \gamma^{\mathbf{a}} z^{-q_{\mathbf{a}}} = 0. \quad (11.2)$$

Fix one exponent vector $\mathbf{a}_0$ in a nonempty residue class. Modulo I ,

$$X^{\mathbf{a}} \equiv z^{-(q_{\mathbf{a}}-q_{\mathbf{a}_0})} \frac{\gamma^{\mathbf{a}}}{\gamma^{\mathbf{a}_0}} X^{\mathbf{a}_0} \quad (\rho_{\mathbf{a}} = \rho_{\mathbf{a}_0}).$$

Therefore the part of P in this residue class is congruent modulo I to

$$\frac{z^{q_{\mathbf{a}_0}} X^{\mathbf{a}_0}}{\gamma^{\mathbf{a}_0}} \left(\sum_{\rho_{\mathbf{a}}=\rho_{\mathbf{a}_0}} c_{\mathbf{a}}(z) \gamma^{\mathbf{a}} z^{-q_{\mathbf{a}}} \right),$$

which vanishes by (11.2). Summing over the residue classes gives $P \in I$. $\square$

For the complete packet, the general binomial ideal has a particularly small set of generators.

Corollary 11.2 (Complete fixed-denominator packet). *For*

$$G_m(z) = \Gamma(m/N)z^{-m/N}, \quad 1 \leq m < N,$$

put $\gamma_m = \Gamma(m/N)$. The kernel of

$$C_{\Gamma,N}(z)[X_1, \dots, X_{N-1}] \longrightarrow \mathcal{M}(\widetilde{\mathbb{G}_m}), \quad X_m \longmapsto G_m(z),$$

where

$$C_{\Gamma,N} = \overline{\mathbb{Q}}(\gamma_1, \dots, \gamma_{N-1}),$$

is generated by

$$\gamma_1^m X_m - \gamma_m X_1^m \quad (2 \leq m < N)$$

and

$$X_1^N - \gamma_1^N z^{-1}.$$

Proof. The first family eliminates every X_m in favor of X_1 . The resulting quotient is

$$C_{\Gamma,N}(z)[X_1]/(X_1^N - \gamma_1^N z^{-1}).$$

The map $X_1 \mapsto \gamma_1 t$ embeds this quotient into $C_{\Gamma,N}(t)$ because $1, t, \dots, t^{N-1}$ are linearly independent over $C_{\Gamma,N}(z)$. Hence the displayed relations generate the full kernel. $\square$

For example, when $N = 3$,

$$G_1(z) = \Gamma(1/3)z^{-1/3}, \quad G_2(z) = \Gamma(2/3)z^{-2/3},$$

and the complete functional ideal is generated by

$$\Gamma(1/3)^2 G_2(z) - \Gamma(2/3) G_1(z)^2$$

and

$$G_1(z)^3 - \Gamma(1/3)^3 z^{-1}.$$

Remark 11.3 (Relation with the Rohrlich–Lang conjectures). Rohrlich’s conjecture predicts that the multiplicative algebraic relations among special rational Gamma values and $2\pi i$ are generated by the standard translation, reflection, and multiplication identities. Lang’s strengthening makes the corresponding prediction for all polynomial algebraic relations [1, §1.4]; see also [36]. Koblitz and Ogus proved the algebraicity implication in the Deligne–Koblitz–Ogus criterion for Gamma monomials; the conjectural converse is a form of Rohrlich’s conjecture [29].

Theorem 11.1 is a relative functional statement, not a result on those constant-value conjectures. Its coefficient field C_Γ is the actual field generated by the chosen Gamma constants, so all relations among those constants have already been imposed. Before extending scalars from $\overline{\mathbb{Q}}$ to C_Γ , the same residue-class argument reduces the unknown part of the functional kernel to linear relations among the Gamma monomials $\gamma^{\mathbf{a}}$. Thus the functional variable contributes only the Kummer-weight binomials above; the remaining arithmetic problem is the relation problem among the chosen Gamma constants, together with $2\pi i$ in the usual global formulation. In this sense, the theorem gives a fixed-denominator functional analogue of the Rohrlich–Lang picture while leaving its classical numerical core open.

11.3 Nilsson– E packets

A finite packet of functions

$$z^\alpha (\log z)^m F(z), \quad \alpha \in \mathbb{Q},$$

with F an E -function has finite Kummer monodromy and, when logarithmic factors occur, possibly infinite unipotent monodromy at the regular singularity. Monodromy projectors and finite differences provide a concrete split model for separating the Kummer and logarithmic components. For a packet placed in the relative exponential Nori envelope, the generic functional theorem applies. Relations at a fixed algebraic value with period-valued coefficients remain arithmetic and are not consequences of the functional theorem.

12 Concluding perspective

The relative exponential Nori category admits the same functional-period structure as the ordinary motivic setting. The fixed-part theorem identifies constant exponential submotives, while the Fourier nearby-cycle and 1-motivic arguments control differential characters. Together they give the relative Galois exact sequence and identify the generic comparison torsor with the Picard–Vessiot torsor.

For the filtered generic Kontsevich–Zagier exponential envelope, the formal functional period map is therefore injective at every algebraic nonzero Fourier scale. The cyclotomic Gamma calculation gives an explicit instance: over the field generated by the constant Gamma values, the complete functional ideal is the Kummer binomial ideal of Theorem 11.1, while the classical Rohrlich–Lang problem remains entirely in the constants. Arithmetic relations created by specialization and an elementary presentation of the general motivic relation algebra are separate problems.

AI-assisted research disclosure

The authors used AI-assisted tools for literature retrieval, proof development, adversarial checking, and document preparation, and are responsible for the manuscript. The AI systems are not authors. The repository records the frozen pre-repair baseline, standalone replacement arguments, and the subsequent assembled-proof and integration audit. Those checks are not a claim of independent human peer review or formal verification.

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